

Path integration as a form of home vector navigation with different intensity of visual, artificial polarization pattern cues in walking bumblebees

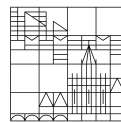
Bachelor's thesis

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Zusammenfassung in deutscher Version

Die Ortung des eigenen Standorts und die Fähigkeit, sich in Richtung einer Nahrungsquelle und zurück zu ihrem Zuhause zu orientieren, sind für jedes Tier grundlegende Prinzipien, die über das Überleben in ihrem Umfeld bestimmen. Das Wegintegrationsverhalten ist eine bestimmte Orientierungsweise, welche von einigen Lebewesen genutzt wird. Dabei wird ein Vektor für den Heimweg mit Hilfe externer Umwelt- oder interner Hinweise, wie zum Beispiel der Anzahl der zurückgelegten Schritte, erzeugt, welcher den direktesten Weg zurück zur Ausgangsposition angibt. Diese Strecke muss dafür nicht vorher zurückgelegt werden.

Für Hummeln ist dieses Verhalten existenziell, da sie täglich mehrere weite Strecken zur Futtersuche zurücklegen und das Überleben der Kolonie von diesen kleinen Lebewesen abhängt. Dabei stützen sie sich unter anderem auf das Polarisationsmuster des Himmels, welches durch die Streuung des Sonnenlichts verursacht wird. Der dorsale Randbereich, ein spezieller Bereich an der Oberseite des Facettenauges, ist für die Polarisationsempfindlichkeit erforderlich. Da das Wetter sehr wechselhaft sein kann und somit auch Wolken und Nebel auftauchen können, kann die Intensität des vorhandenen Polarisationsmuster stark variieren und der Polarisationsgrad abnehmen. Daher stellt sich die Frage, ob Hummeln in der Lage sind, die Wegintegration bei unterschiedlicher Intensität vorhandener Polarisationsmuster durchzuführen, wenn sie nur diesen Stimulus zur Orientierung zur Verfügung haben.

In dem hier durchgeführten Experiment ermöglichte eine Arena mit Zugang zu Futter in der Mitte, in der polarisiertes Licht ausgestrahlt wurde und kein anderer Stimulus gegeben wurde, den Hummeln die Futtersuche in einer beobachteten Umgebung und führte zur Beantwortung dieser Frage. Die Ergebnisse zeigten, dass die Hummeln mit abnehmender Intensität der Polarisationsinformation immer noch in der Lage sind, ihr Nest zu finden. Dies kann daran liegen, dass die Hummeln nicht nur die vorhandenen externen Informationen zur Orientierung benutzen, sondern auch ihr Langzeitgedächtnis zur Hilfe nehmen. Dabei generieren sie pfadabhängige Vektoren während der Erkundung des Netzes und merken sich diese. Die Hummeln orientieren sich dann mit Hilfe ihres Gedächtnisses zurück zum Nest, wobei sie Informationen aus ihrem Kurzzeitgedächtnis mit ihrem Langzeitgedächtnis kombinieren. Dies würde für eine Lösung der Orientierung beim Auftreten von Wolken oder Nebel sprechen.

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ABSTRACT

The knowledge of location and the ability to orientate towards a food source or back to their nest or home are fundamental for any animal. Therefore, these animals perform certain types of behaviors to fulfill this task. The ability to generate a home path vector indicating the most direct path back to their initial position with the help of any external environmental or internal cues, as the number of steps, is termed path integration behavior.

For bumblebees, it is even more important to perform path integration, as they perform multiple foraging paths a day and thus, the survival of the colony depends on these small foragers. To do so, they rely, amongst other cues, on the skylight polarization pattern, which is caused by scattering of sunlight. The dorsal rim area, a specialized area on the upper side of the compound eye, is required for polarization sensitivity. As the weather changes frequently with the appearance of clouds or fog, the intensity of the polarization pattern is not always equal and the degree of polarization decreases. Consequently, the topic of the bumblebee's capacity to navigate under varying levels of polarization patterns emerges.

An arena with a feeder located in the middle, in which polarized light was emitted and no other cues were provided, allowed the bumblebees to forage under an observed environment and provided answers to this question. With decreasing intensity of polarization information, the bumblebees are still able to find their nest. The flexible usage of their long-term memory and therefore the generating of path-indicated vectors, is implied. As the bumblebees become more familiar with the constructed arena, they are able to memorize their environment. The bumblebees then orientate back to the nest with the help of their memory, combining information of their short-term memory with their long-term memory. This would suggest a solution in an environment with no available polarization cues, as this could occur with the appearance of clouds.

1 INTRODUCTION

1.1 PATH INTEGRATION

The world is dominated by visual sensations. These visual cues are used by animals to understand their location (Brines and Gould, 1982; Greif et al., 2014). Orientating in animals is fundamental to survival, underscoring the importance of effective strategies for navigating the environment (Heinze et al., 2018). Some animals, including insects, rodents, and humans, have the ability to continuously track their position relative to a defined starting point (Heinze et al., 2018; Savelli and Knierim, 2019). They then integrate the information about the directions and distances they have traveled with the help of external landmarks to transform it into a sense of location (Heinze et al., 2018; Savelli and Knierim, 2019). Therefore, the transformation of this information results in a most direct “home vector” which directly points back to the initial starting point (Heinze et al., 2018; Patel et al., 2022), as shown as in Figure 1 (Egelhaaf and Lindemann, 2025). The animal’s previous experience with this home path is not relevant. This vector-based navigation is called path integration (Patel et al., 2024).

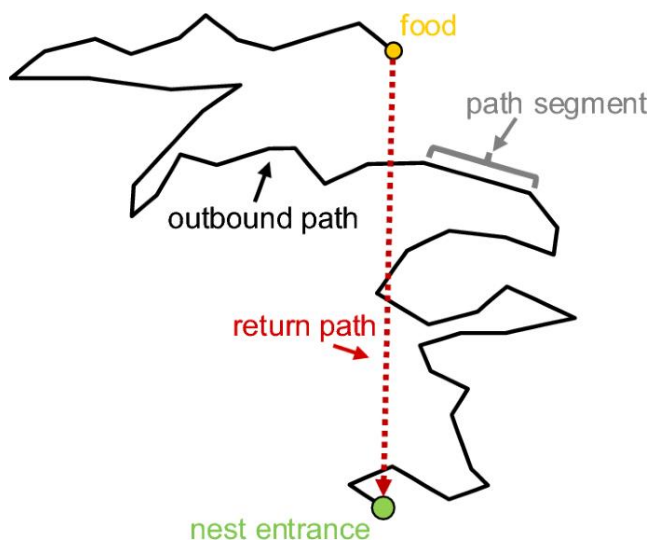


Figure 1 - The schematic and basic principles of path integration

The animal is walking from its nest entrance, which is its defined starting point, to an unknown food source. Therefore, it takes an outbound path by walking in many small path segments or vectors, each pointing in a specific direction, searching for the food source. As the animal arrives at the food source, the many vectors will be added to a home vector, which indicates the most direct return path back to the nest

Reproduced from Egelhaaf and Lindemann (2025)

During the Central place foraging behavior of some animals, such as bumblebees, it is important for them to return to their homes and bring the food to their nest after traversing various foraging paths to a food source (Ylitalo et al., 2021). They make use of external (allothetic) cues, such as visual objects and polarization patterns, to orient themselves (Heinze et al., 2018; Patel et al., 2022). They also rely on self-generated movement (idiothetic) cues, including the number of steps taken, to estimate distance (Heinze et al., 2018; Patel et al., 2022).

For this study, bumblebees (*Bombus spp.*) are appropriate as these animals are highly motivated to do foraging (Patel et al., 2022) and therefore, their home path can be observed repeatedly. Furthermore, the maintenance of a bumblebees colony is controllable (Patel et al., 2022). Bumblebees are characterized by their eusocial structure, which is typified by a division of labor within their yearly colonies (Ge et al., 2023; Wilson, 1976; Wilson, 1976). The queen's primary responsibilities include egg-laying and the maintenance of the brood in the nest (Goulson, 2009), which is often constructed underground in open uncultivated locations (Kells and Goulson, 2003; Svensson et al., 2000). In the early stages of a novel colony, newly emerged female bumblebees engage in foraging activities to collect pollen and nectar, or alternatively, they allocate their efforts towards brood care (Goulson, 2009; Ratnieks and Wenseleers, 2008). Subsequently, the development of male bumblebees commences, thereby initiating the reproductive phase and the beginning of a new colony (Amsalem et al., 2022; Goulson, 2009).

In summary, orientation is important for every individual, especially for bumblebees, which forage multiple times per day. To orientate and navigate, they perform path integration and combine allothetic cues, such as the skylight polarization patterns, with idiothetic cues to create a home vector straight back to their nest.

1.1.1 “vector map” or “cognitive map” – Short- and long-term memory of bumblebees

The information used for path integration behavior, can be stored in the animals' spatial memory (Balda et al., 1998; Menzel et al., 2000). Honeybees are able to recall their surroundings as novel shortcuts between locations of their foraging trips (Menzel et al., 2005). The capacity to identify new, more direct paths between various locations, frequently by integrating learned pathways or using environmental cues, is referred to as novel shortcuts. (Foo et al., 2007; Foo et al., 2010; Patel et al., 2024). The utilization

of these novel shortcuts enables the design of optimal, direct routes to multiple locations (Lihoreau et al., 2012; Ohashi et al., 2007). The formation of novel shortcuts could be provided by the short-term memory (Menzel et al., 2000; Patel et al., 2024). Additionally, the recall of novel shortcuts can be facilitated by a combination of long short-term memory, which indicates a more advanced method (Schlichting et al., 2022). While it has not been demonstrated conclusively in bumblebees, the behavioral and neural mechanisms observed in honeybees suggest a similar memory-based process is likely.

Furthermore, Patel and colleagues (Patel et al., 2024) have proposed that, in the event that home vectors are saved and linked to well-known locations, an associated home vector can be accessed once a familiar location is identified. They suggest that this vector can then be used for navigation back to one's place of origin or to another location, depending on where the vector points. Additionally, if the bumblebee seeks to traverse from one known location to another, they argue that the bumblebee has the ability to integrate the memorized home vectors from both locations to formulate a new vector that will guide it between these destinations. Patel and colleagues imply that a "vector map" is generated by the transfer of multiple vectors to parallel long-term memories, thereby obviating the necessity for a centralized, classical cognitive map (Behrens et al., 2018). Cognitive maps are internal representation and allow individuals to picture relations between different objects and locations, which is useful for their orientation and spatial navigation to move freely in an environment (Behrens et al., 2018; Epstein et al., 2017; Park et al., 2020). Thus, a flexible navigation is possible (Behrens et al., 2018).

Patel and colleagues' (Patel et al., 2024) study supports the hypothesis that the short-term and long-term mechanisms of bumblebees can be investigated by observing their home path behavior in response to the rotation of a stimulus cue of linearly polarized light. They argue that, in the event that the bumblebee does not follow the stimulus direction, instead using hive-directed vectors, they are using their long-term memory.

The arguments put forward by Patel and others would suggest that bumblebees likely rely on both short-term and long-term memory to perform flexible, goal-directed navigation, which can be experimentally dissociated through manipulations of the angle of polarized light.

1.2 POLARIZATION

Animals have the ability to rely on a variety of cues for path integration. However, given the significance of the celestial cue and its high degree of availability, the utilization of skylight patterns has become a common cue for animals to rely on (El Jundi et al., 2015; Horváth, 2014; Nguyen et al., 2022; Patel et al., 2022). Linearly polarized light, produced by a variety of natural sources, is a constant presence in the environment experienced by flying insects (Horváth, 2004). This celestial cue plays a fundamental role in their capacity for navigation and migration (Reppert et al., 2016), communication (Dusenbery, 1992), and resource location (Horváth, 2004), including water and prey (Johnsen, 2012).

In classical optics, light can be thought of waves of electric and magnetic fields (Foster et al., 2018). Their electromagnetic fields oscillates perpendicular to the direction of travel (Cronin, 2014). As polarization is the average orientation of the electric field vector of a light wave (Born and Wolf, 1999; Gmelch and Reineke, 2019), polarization depends on the direction in which these fields are oscillating (Johnsen, 2012).

Most light sources, such as light bulbs, candles, and the sun, produce light that is composed of electromagnetic waves that are orientated in all different directions (Hecht, 2002). Therefore, it is unpolarized light or, at most, weakly polarized (Cronin and Marshall, 2011). Because of air molecules, aerosol particles and clouds in the atmosphere (Li et al., 2024), polarized light is produced by Rayleigh scattering (Cronin and Marshall, 2011). Polarization could also occur because of reflection at water surfaces, at the surfaces of wet objects or at the surfaces of some leaves or insect cuticles (Cronin and Marshall, 2011).

Some materials selectively block electromagnetic waves orientated in a specific direction, while allowing waves polarized perpendicularly, to pass through (Hecht, 2002; Johnsen, 2012). These optical filter materials may occur in nature, such as chitin structures in insect cuticles or the photoreceptors in the dorsal rim areas of bees (Horváth, 2004; Johnsen, 2012; Schinz, 1975), or in artificial polarization filters such as the Polaroid™ filters or dichroic filters used in cameras, sunglasses or in microscopy (Hecht, 2002; Johnsen, 2012; Pedrotti et al., 2018). The filter materials could also slow down the speed of the light waves (Born and Wolf, 1999). Depending on the effective speed each wave orientation travels at, different types of polarization, such as linear,

circular, and elliptical, are produced.(Hecht, 2002; Johnsen, 2012) They exhibit distinct characteristics and behaviors. (Sparks et al., 2009)

However, the polarization types utilized in the experiment were limited to unpolarized and linearly polarized light. Therefore, only unpolarized and linearly polarized light are described. For the sake of simplicity, the text only mentions polarized light when further describing the linearly polarized light used in the experiment.

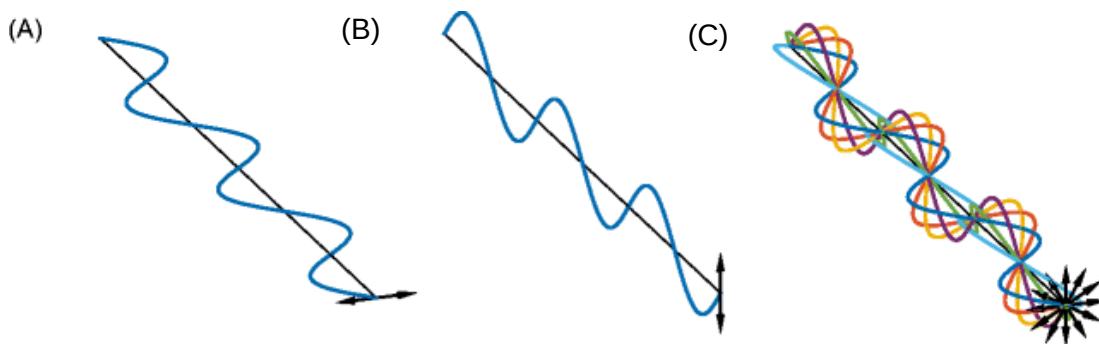


Figure 2 – Schematic representation of two –unpolarized and linearly polarized - polarization types

A, B) Two different sates of linearly polarized light are represented. In A), horizontally linearly polarized light is illustrated. B) represents vertically linearly polarized light. The electric field vector oscillates in one single direction that lies in a plane perpendicular to the direction of motion.

C) In unpolarized light, the beam consist of many waves with an uniform distribution of angles

Adapted from Foster et al. (2018)

1.2.1 Unpolarized light

Unpolarized light does not exhibit a consistent electric field orientation over time (Foster et al., 2018). While each individual wave within the beam still oscillates in only one single direction, the beam as a whole consist of many waves with a uniform distribution (Foster et al., 2018). The light is considered completely unpolarized, when all three polarization-related Stokes parameters are equal, as they describe the state and the total intensity of polarization (Born and Wolf, 1999; Kozlov et al., 2018). It further implies the absence of any net or dominant polarization state (Kozlov et al., 2018; Kroh et al., 2020).

1.2.2 Linearly polarized light

Linearly polarized light occurs when the electric field vector oscillates in a single plane perpendicular to the direction of the light's propagation (Manion and Stokkermans, 2025). Thus, the electric field vector or the axis of the electrical field of the wave will be constant (Cronin, 2014). Depending on the angle the electric field vector makes,

linearly polarized light is described as vertically polarized, horizontally polarized, or any angle in between (Cronin, 2014; Foster et al., 2018).

In the case of a linear polarized light filter, irrespective of the form of light entering the material- be it circularly polarized, linearly polarized, or unpolarized – only linearly polarized light is capable of existing the material (Hecht, 2002; Johnsen, 2012; Pedrotti et al., 2018). The filter selectively absorbs light waves which are oscillating perpendicular to that direction (Sedov et al., 2019) as illustrated in Figure 3.

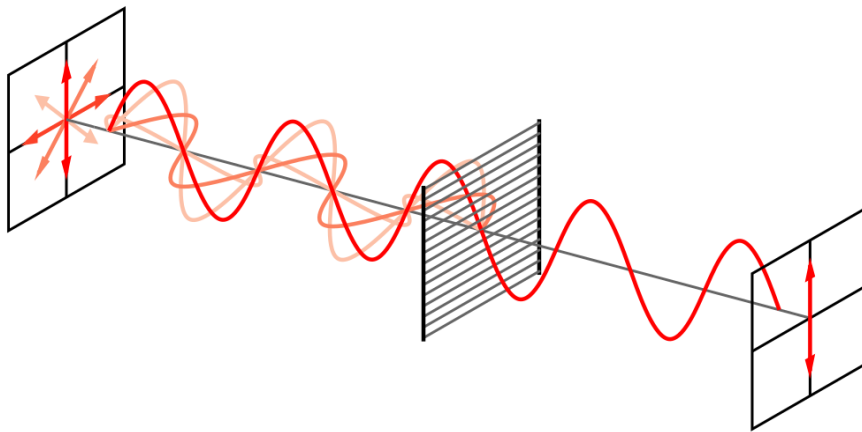


Figure 3 – A filter material, as a linear polarizer, transforms unpolarized light into linearly polarized light

The filter absorbs light waves oscillating horizontal, while it transmits the waves oscillating vertical. Therefore, only vertical linearly polarized light is able to pass through the polarizer. The unpolarized light is converted to linearly polarized light.

Reproduced from Wikipedia.org,, uploaded 21.04.2006, illustrated by Bob Mellish

<https://en.wikipedia.org/wiki/Polarizer#/media/File:Wire-grid-polarizer.svg>

If the incoming light is already linearly polarized, and if it is oriented in the direction, in which the material blocks all electromagnetic waves, then it can be concluded that all of the light will be blocked completely (Collett, 1993; Hecht, 2002; Johnsen, 2012).

In summary, animals frequently use celestial cues for navigation. Specifically, linearly polarized light from skylight scattering functions as a reliable and widespread signal for path integration, orientation, and resource detection.

1.3 POLARIZED-LIGHT PATTERNS IN THE SKY WITH SUNLIGHT

Skylight polarization patterns, a naturally occurring phenomenon caused by scattering of sunlight in the atmosphere, have been shown to serve as navigation cues for certain species (Homberg et al., 2011; Horváth, 2014; Wehner, 2001). In Guan's study, they argue that the patterns provide directional information (Guan et al., 2018). Furthermore, they suggest that the information it provides pertains to the solar meridian, the neutral point, and the polarization angle. The solar meridian is defined

as an imaginary line through the sun's position and the zenith (Muller et al., 2017). Thus, it divides the celestial sphere into eastern and western regions (Tate, 2010). The neutral point lacks emission of polarized light in specific areas (Horváth, 2004; Horváth et al., 2002). Consequently, the position of the sun, or the solar azimuth, could be detected by the animal and utilized for navigational purposes (Foster et al., 2018).

Important for its definition, the degree of polarization (DOP) and the angle of polarization (AOP) have to be explained. AOP refers to the axis along which the predominant proportion of a light wave's electric field strength is oriented (Foster et al., 2019), whereas the DOP is the percentage of polarized light intensity in total light intensity (Kong et al., 2023).

As the Figure 4 shows, the DOP of skylight exhibits a minimum close to the sun and anti-sun (Pfeiffer et al., 2011), where the neutral points (Horváth et al., 2002) with a DOP of 0 occurs (Cronin and Marshall, 2011). A band of maximum degree of polarization is observed at angles near 90° from the sun (Cronin and Marshall, 2011). Due to the hemispherical shape of the observable sky, this polarized light is mostly observed in the anti-solar half of the sky (Foster, 2024). A gradual rise in the DOP, commencing at a proximity to the sun and anti-sun, and culminating at a 90° to the sun, is evident (Cronin and Marshall, 2011).

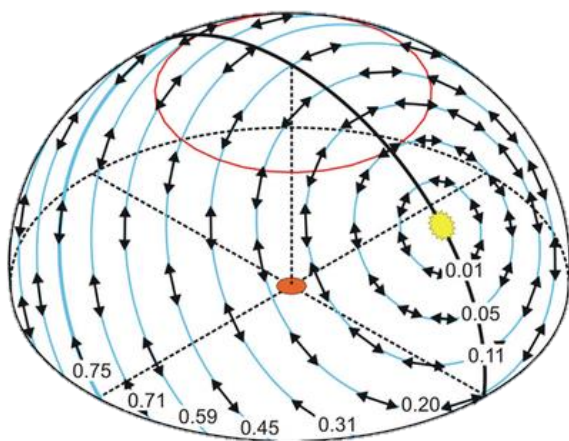


Figure 4 – Polarized-light patterns of skylight

Observing it from the center of the sphere (red spot), the sun is at a solar elevation of 40° , indicated by the yellow point. The blue lines demonstrate the degree of polarization with a maximum of 0.75. The light close to the sun is weakly polarized with a DOP of 0, whereas the highest degree of polarization is observed along the solar equator at angles near 90° to the sun. Adapted from Pfeiffer et al. (2011)

Therefore, a certain particular of angles of polarization, as represent in Figure 4, occurs, allowing the animals to use these polarization patterns to determine their orientation (Homberg et al., 2011).

1.3.1 Clear sky

In the condition of a very clear sky with bright sunlight and an absence of clouds (Figure 5), the DOP of the band at angles near 90° from the sun could approach 90 per cent (Cronin and Marshall, 2011). However, in most cases, it will be lower than 90 per cent (Cronin and Marshall, 2011).

In the instance that the sun is positioned in close proximity to the horizon, as is the case during the dim light of dawn and dusk, the maximum of the DOP of the polarization pattern of scattered light from celestial sources is reached (Araújo et al., 2024).

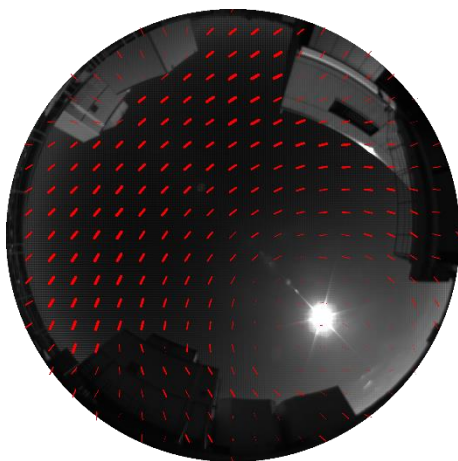


Figure 5 – The polarization pattern in a bright and clear sky with the sun shining

The red lines show the AOP, and the thickness of the lines indicates the DOP. Close to the sun, the lines are disappearing as the DOP reaches its minimum close to the sun. The images were taken with a fisheye polarization camera and processed with Python

Figures courtesy of Georgios Kolyfets

1.3.2 Cloudy sky

In the presence of clouds, an increased quantity of suspended particles is present in the atmosphere (Isokääntä et al., 2022; Weigelt et al., 2009). Consequently, the effect of multiple-path scattering is amplified, resulting in a depolarization of the light (Cronin and Marshall, 2011). The DOP decreases (Wang et al., 2016), as shown in the three

different cloudy conditions in Figure 6. A similar effect is also observed for air pollution in the lower atmosphere (Cho et al., 2021).

In addition, the findings of Aycock et al.'s experiment demonstrated, that the precision of polarization azimuth is reduced in overcast conditions relative to clear skies (Kong et al., 2023). In conditions of cloud cover, the DOP reaches approximately 40% at angles of 90° close to the sun (Können et al., 1985).

Nevertheless, given the extensive array of cloud types and the numerous factors that influence the DOP of cloud light, the DOP may exhibit variability among different cloud formations (Pomozi et al., 2001).

Nonetheless, should there be regions of the sky that remain uncovered by clouds, animals that navigate using polarization pattern cues may be able to utilize these patches of visible sky (Cronin and Marshall, 2011). The minimum DOP, that is essential for ensuring accurate navigation within a specific patch of skylight is determined to be approximately 5-10% (Pomozi et al., 2001).

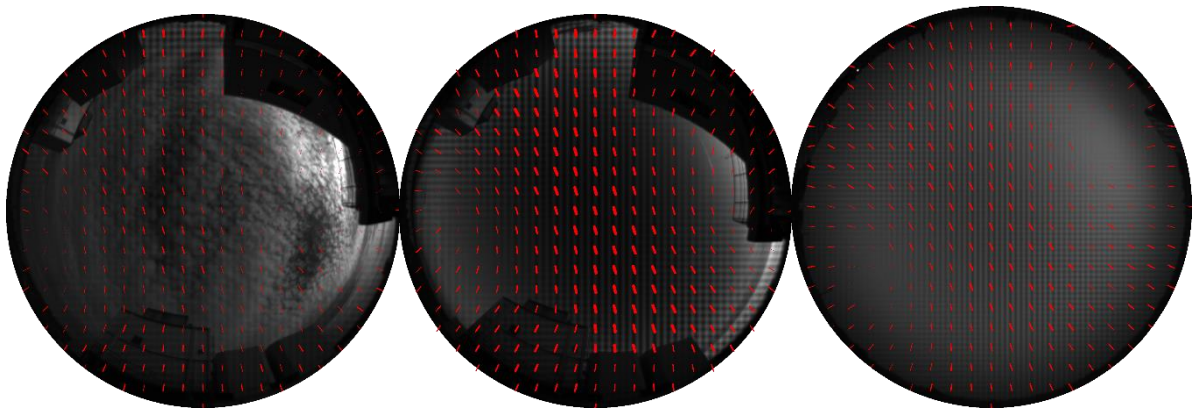


Figure 6 – The polarization patterns in three different cloudy conditions

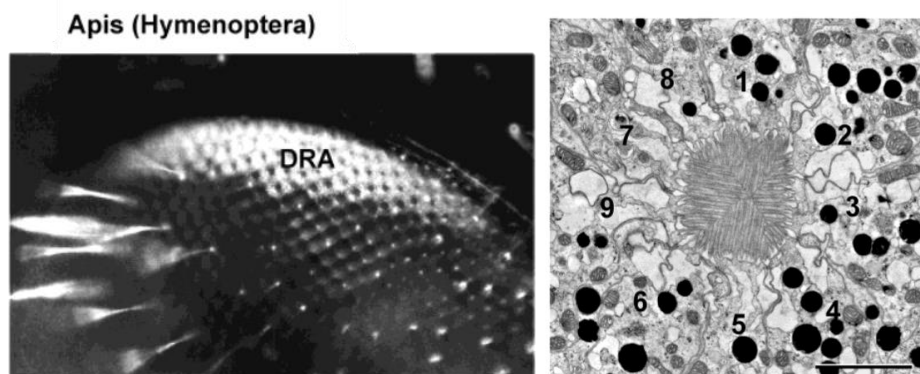
The red lines show the AOP, and the thickness of the lines indicates the DOP. In cloudy conditions, the DOP decrease because of more scattering by suspended particles in the cloud. Depending whether there are still some patches of clear sky, the DOP can still reach higher values of DOP. The images were taken with a fisheye polarization camera and processed with Python. Figures courtesy of Georgios Kolyfets

Therefore, in comparison to clear sky, cloud cover increases the scattering in the atmosphere, which leads to an increase in the degree of polarization and the precision of polarization cues. While the degree of polarization can vary with the diversity of cloud types, small clear patches of the sky can still provide enough information about the polarization for accurate navigation.

1.4 DORSAL RIM AREA – POLARIZATION SENSITIVITY IN BUMBLEBEES

The capacity of animals to utilize polarization pattern cues of the skylight for navigation is dependent on the presence of specialized visual systems and receptors, which facilitate the detection and processing of this information (Greif et al., 2014; Ma et al., 2015; Wehner, 2001).

In the eyes of bumblebees, honey bees, locusts, desert ants, fruit flies and many other insect species (Labhart and Meyer, 1999), a specific eye region is essential for polarization sensitivity (Hardcastle et al.; Labhart and Meyer, 1999; Zittrell et al., 2020). The rhabdoms are photosensitive structures located in the eyes of many invertebrates (Grusch et al., 1997; Reisenman et al., 2002). The rhabdoms contain photosensitive cells that are characterized by their dense microvilli (Cronin, 2014; Rutherford and Horridge, 1965), which play a fundamental role for light absorption and phototransduction (Eguchi et al., 1962; Rutherford and Horridge, 1965). The specific configuration of these microvilli determines a certain function of the rhabdoms (Foster, 2024; Greiner et al., 2007; Matsushita et al., 2012). In the dorsal rim area (DRA) of the compound eye (Araújo et al., 2024; Foster, 2024), the rhabdoms play a crucial role for polarization sensitivity (Foster, 2024; Greiner et al., 2007; Karman et al., 2012). The DRA is positioned in the upper portion of the compound eye and exhibits a distinct structural configuration compared to regions in the compound eye outside the DRA (Jie et al., 2023). The DRA is lighter in color compared to the surrounding regions of the eye, noticeable as a grey area within the compound eye, which, as shown in Figure 7 (Labhart and Meyer, 1999). This gray coloring is associated with its ability to disperse light (Foster, 2024).



*Figure 7 – Representation of the location of the dorsal rim area and an electron micrograph of cross-sections through the dorsal rim area of the honeybee *Apis mellifera* (Labhart and Meyer, 1999)*

The dorsal rim area is detectable as a light-grey area at the upper part of the compound eye. In the right picture, nine receptor cells, instead of eight as in the rest of the compound eye, are visible. Their microvilli are organized orthogonally to each other. Adapted from Labhart and Meyer (1999)

In contrast to the characteristic twisting of microvilli observed in other types of rhabdoms in the bee compound eye, the photoreceptor cells of *Bombus terrestris* exhibit a distinct absence of such twisting (Jie et al., 2023; Meyer-Rochow, 2019). Consequently, the microvilli are arranged in a parallel, untwisted configuration along the length of a specific photoreceptor (Foster, 2024; Wehner et al., 1975). The formation of untwisted and highly ordered microvilli allows each photoreceptor to respond most strongly to a specific angle of polarized light (Greiner et al., 2007; Karman et al., 2012). Therefore, the bumblebees are able to detect the direction of polarization (Foster et al., 2014; Homberg et al., 2011), and therefore the electric field vector orientation (Brines and Gould, 1982), which is essential for using the information as a navigation strategy (Greiner et al., 2007). The DRA is therefore highly sensitive to the angle of polarized light (Homberg and Paech, 2002) as the organization of the microvilli leads to a very strong absorption (Cronin, 2014).

Moreover, electrophysiological studies demonstrate that the DRA photoreceptor can be coupled in bumblebees, which leads to a spatial summation that expands the receptive fields of the neurons, thereby enhancing their sensitivity to broader patterns (Kolyfetis et al., 2025).

In numerous insects, the DRA is instrumental in the polarization sensitivity. (Wehner et al., 1975) Additionally, its dimensions and complex configuration may exhibit variability based on the species and the subject's body size. (Jie et al., 2023) Therefore, it adjusts to the unique characteristics with which it is required for the use of the sky polarization pattern as a compass, supporting orientation, navigation and foraging behaviors (Foster et al., 2019; Sparks et al., 2009; Stalleicken et al., 2006).

In summary, bumblebees detect polarized light through specialized photoreceptors located in the dorsal rim area of their compound eyes. The untwisted and linearly ordered microvilli in the dorsal rim area make them strongly sensitive to the angle of polarized light, which allows the bumblebees to use skylight polarization patterns for their location and navigation.

1.5 AIM OF THE STUDY

The aim of the present study is to examine the navigation behavior exhibited by bumblebees in response to a polarization stimulus that has been provided. The central hypothesis posits that the observed navigation behavior of the bumblebees is a response to the polarization stimulus and if this behavior is affected by varying degree of polarization. A secondary research aim is to determine whether this behavior is affected by varying degrees of polarization. This would be analogous to the skylight polarization pattern under different weather conditions. The subsequent hypothesis posits that the bumblebees will demonstrate a reduced deviation in their orientation towards the nest under specific degrees of polarization. Furthermore, an investigation into the utilization of an alternative orientation method by the bumblebees is conducted, as it would be observed in an environment with the appearance of cloud, thereby leading to a reduction in degree of polarization. The use of a flexible combination of short- and long-term memory is suggested. The investigation of these hypothesis involves varying the degree of polarization and the orientation of the stimulus.

2 EXPERIMENTAL DESIGN

2.1 LABORATORY ANIMAL

In order to conduct the experiment, bumblebees of the species *Bombus terrestris* were first required.

These bumblebees were bred by the company Biobest Deutschland GmbH, and were delivered in a box with a special nutrient-rich sucrose solution and pollen. Prior to the arrival of the bumblebees, the boxes used for the experiment were sterilized by torching them with a blow torch. Subsequently, the boxes were filled with cat litter. The boxes were then sealed at the top with Plexiglas and tape (TESA, PERFECT fabric adhesive tape, 56339-00002-01).

After delivery, the entire colony, which initially comprised between 30 and 50 individuals, was transferred with the nest structure into a nest box, which was situated within a cooling chamber at the University of Constance. The climate chamber maintained a precisely controlled environmental temperature range of 25°C to 30°C and a humidity level between 15% and 60%, thus providing a stable and controlled environment for the study's bumblebees. The optimal temperature range within the nest was from 30°C to 35°C, which the bumblebees further achieved by thermogenesis (Weidenmüller et al., 2022). The temperature was recorded using a thermometer (TFA Dostmann radio thermometer, 30.3065.02, Conrad Electronic). The lighting in the cooling chamber was set to a 12:12-hour light-dark cycle, commencing at 10:00 a.m.

In the course of this experiment, the walking behavior of the bumblebees was investigated. For this purpose, their wings were clipped one to two days after their arrival. In order to accomplish this task, the bumblebees were initially anesthetized with carbon dioxide, which was supplied via a prepared soda stream. The wings were clipped at the base using scissors. During the anesthesia, the bumblebees were also marked with markers (Opalith Classic & Leucht marker plates, different colors), which were adhered to the top of the thorax. To facilitate the distinction of individuals, the bumblebees were assigned distinct numerical identifiers in various colors. This process was repeated whenever new individuals hatched within the colony.

Upon arrival, the bumblebees were provided with access to a 0.5M sucrose solution (35g of sugar cube Südzucker + 200ml of H₂O) during the preparation phase via a

gravity feeder within a feeding box. During this period, the access to the arena remained completely closed. In the initial days following their arrival, the 0.5M sucrose solution was further enriched with the delivered nutrient-rich food source to support the bumblebees. The gravity feeder was equipped with multiple holes at its base, allowing the bumblebees to access the sucrose solution. The sucrose solution was refilled daily and freshly prepared as needed. At the end of each week, a minimal quantity of pollen was administered and dispersed into the nest box. The pollen ration was maintained at a low level, as the cultivation of a small colony size was considered optimal.

To ensure the well-being of the bumblebee colony, the arena and the connecting tubes were cleaned regularly using a cleaning brush (Flexible Pipe Cleaning Brush, Lejie) and a 70% ethanol solution (70 mL of ethanol and 30 mL of H₂O). Furthermore, the presence of deceased bumblebees was documented and the bodies were subsequently removed, thereby reducing the risk of microbial diseases within the colony. The removal protocol for dead individuals involved placing them in a freezer, followed by proper disposal via specialized animal waste procedures. A daily assessment was conducted to monitor the quantity of food consumed by the bumblebees and the amount of honey stored in the nests. In the event that the honey pots were found to be insufficiently filled, sucrose solution was manually added to the pots.

Upon transitioning from the preparatory phase to the training phase, access to the feeder box was restricted, while access to the arena was opened. Within the arena, the bumblebees were able to access a food source via a conical feeder (see experimental preparation).

Two distinct colonies were examined over the course of the experiment, which spanned six weeks.

2.2 SETUP OF THE EXPERIMENTAL APPARATUS

As previously stated, the bumblebee colony and its nest were transferred to a prepared nest box. A perforation was present on one side of the box, to which tubes were affixed, allowing the bumblebee to traverse the distance to the arena and feeder box. These tubes could be closed at various points using small chips, which were used to deny or allow access to the arena, the feeder box, or the nest box. The feeder box contained

two gravity feeders filled with a 0.5M sucrose solution in two slightly separate chambers. The upper portion of each box was designed to be opened.

Access to the arena branched off between the two entry points located at either end of the nest and the feeder box, which is shown in Figure 8. A conical feeder was positioned centrally within the arena, through which the bumblebees were provided with a 1M sucrose solution (equivalent to 70 g of sucrose (Südzucker sugar cubes) and 200 mL of H₂O) via multiple apertures in the floor. The feeder was strategically situated at the base of the arena to ensure it did not serve as a visual stimulus for the bumblebees. The feeder could be filled and positioned at the base of the arena. The distance between the conical feeder and the arena entrance measured 39 centimeters. A round curtain was affixed to the periphery of the arena.

The arena was closed at the top with a polarization device, which was aligned with the gravitational horizontal. Two circular rings, each comprising a UV LED (CUN6GF1A - Seoul Viosys UV-A LED 365nm, CA3535) and a linear polarizing filter in between (Bolder Vision Optik UV, Boulder, Colorado, USA – 1000 x 600 mm Dichroic Film Polarizer Sheet, 300 - 700 nm transmission), were connected to this apparatus, enabling modulation of the intensity of the polarized light. A controller was constructed for adjusting the intensity polarized light, and another controller was built for unpolarized light. Both controllers were equipped with a range of settings, ranging from 0 to 8, allowing for precise adjustment. With higher settings, the LEDs responsible for either unpolarized or polarized light become brighter. This setup generated an area within the arena where the polarized light required for the experiment was emitted from above. The polarization device was capable of being rotated by 90°, thereby enabling the configuration of the "top, bottom" (TB) and "left, right" (LR) settings. In the "TB" condition, the angle of polarization was oriented along the axis extending from the arena entrance to the feeder, thereby aligning with the orientation of the training position. The bumblebee was entering the arena, and moving toward the feeder under the familiar lighting conditions. This position was described as "vertical" in the following text. During the feeding process at the feeder, the stimulus was rotated anticlockwise by 90°. Consequently, the bumblebee traversed back to the nest entrance, exhibiting a shift in the angle of polarization, which was now 90° shifted to the axis from feeder to the nest entrance, which is indicated as the "horizontal" position. In the "LR" condition, the initial angle of polarization was 90° to the path from entrance to feeder

before the bumblebee was entering the arena. Therefore, the rotation of the polarization device during feeding resulted in the vertical axis of the angle of polarization, as in the training. The rotation of the angle of polarization can be used to determine whether the bumblebees react to the provided stimulus and to investigate if the bumblebees are using this cue for their orientation and navigation back to their nest.

The setup of emitted polarization resulted in the identification of two distinct areas that required examination. On the one hand, the area in which the polarized light was emitted, indicated as the inner circle, which was fundamental for the evaluation of the experiment, and on the other hand, the area that was not irradiated, extending up to the curtain, referred to as the outer circle. In the videos selected for analysis, these regions were distinctly delineated by the presence of two dark rings.

The light cycle was configured to a 12:12-hour light-dark cycle, with a light phase from 10:00 a.m. to 10:00 p.m. During the preparatory phase, light (LED-DRIVER, Input: 220-240-, 50/50 Hz, 110mA, PF>0,5, Output:80-95 Vdc, 135mA, max 12W + LED-Board, 230V/ 50 Hz, 90v DC/12W, Art-Nr 994148, Leuchten Direkt GmbH) was emitted throughout the room located outside the designated arena. During the training and experimentation phase, the light source was transitioned to the light produced by the polarization device. During the training phase, the experimental setting utilized polarized light with an intensity of 8 and unpolarized light with an intensity of 0, designated as the “train” (no rotation) condition. The experiment was conducted using a variety of settings.

A camera (ELP Full HD web camera, 2.1 mm, wide-angle lens), situated on the floor beneath the arena, was used to record the different trials of the individual bumblebees. The camera was able to record the trials because the floor of the arena was designed to be transparent. Additionally, the arena was illuminated with an infrared light to create a clear shadow. The documentation of these trials was facilitated by the use of a laptop computer, employing the open source software (Open Broadcaster Software (OBS); <https://obsproject.com/>) for recording and data storage.

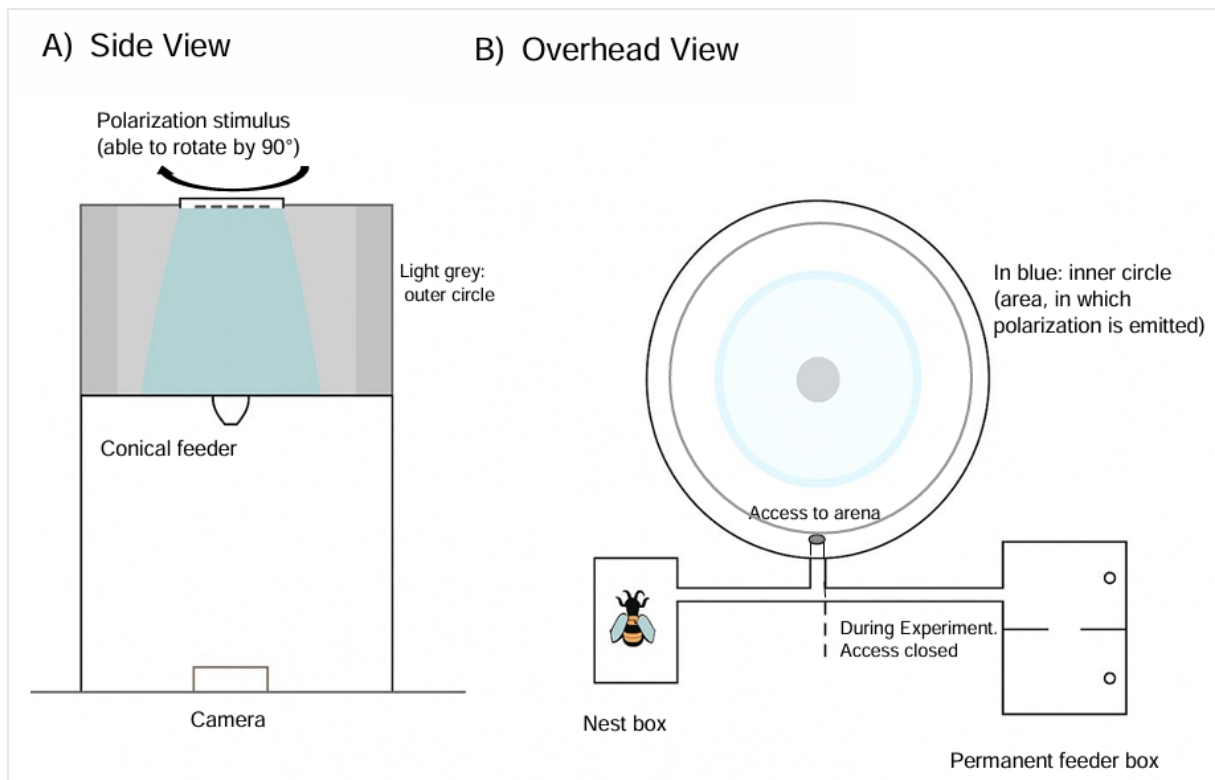


Figure 8 - Experimental design
designed by Marie Jansen, digitalized by ChatGPT 4.0

To avoid disturbing the bumblebees, red LED light (Paulmann SimpLED 78979 LED strip complete set with plug 12 V 7.5m RGB 1 pc, Conrad Electronic) was used when directly engaging with the bumblebees.

2.3 EXPERIMENTAL PREPARATION

The transition from the preparatory stage to the training stage was accompanied by a modification in the light stimulus. The intensity of the polarized light was set to 8, while the intensity of the unpolarized light was set to 0. The bumblebees were prevented from accessing the feeder box, while access to the arena was permitted concurrently. Consequently, the bumblebees had permanent access to the arena, which enabled them to explore it independently.

The objective of the training phase was to instruct foraging behavior in the bumblebees and to support the subsequent localization of the conical feeder within a particular light stimulus setting. To facilitate navigation, a sucrose solution trail was also established from the entrance of the arena to the feeder during the initial days. The duration of the training phase typically ranged from three to seven days, depending on the number of

bumblebees that successfully navigated directly to the feeder and subsequently returned to the arena.

The individual identification of bumblebees deemed potentially suitable for the experiment was recorded in advance.

2.4 EXPERIMENTAL PROCEDURES

The initiation of the experiment was contingent upon the presence of a sufficient number of foraging bumblebees that had memorized the localization of the conical feeder. In order to fulfill this task, it was first necessary to remove all bumblebees from the arena. Consequently, access to the arena and the nest was prohibited by closing the access of the tubes with the small gates. The bumblebees from the arena were extracted with a specialized insect vacuum cleaner (Exbuster Insect Vacuum Cleaner with LED Light, 35cm, NX5747-944) and transported back to their nest via a small transferring box.

In order to eliminate the potential for additional stimuli that might guide bees back to the nest entrance, such as odors, the entire arena was sanitized using a 70% ethanol solution. In the event that a deceased bumblebee was discovered within the arena, it was removed from the arena. The conical feeder was completely refilled.

In each experiment, a single bumblebee was utilized for testing. Therefore, it had to be separated. The use of the gates facilitated the isolation of a single bumblebee from the others within the tubes.

Prior to the release of the bumblebee into the arena, the stimulus setting to be tested was established, and video recording was initiated using the open source software OBS. Upon the arrival of the bumblebee at the feeder, its feeding behavior resulted in a 90° rotation of the angle of polarization. The direction of the rotation depended on the required setting, which was indicated by LR or TB condition. The duration of a trial was determined by the distance traversed by the bumblebee from the entrance of the arena to the feeder until it exited through the curtain at one point.

After exiting at one point of the curtain, the video was stopped, and the bumblebee was transported back to its nest via a transferring box with the assistance of the insect vacuum cleaner.

Subsequent to each trial, the arena was cleaned using a 70% ethanol solution.

2.5 SETTINGS AND CONDITIONS

In order to ascertain the suitability of the bumblebees recorded in advance for the experiment, it was first necessary to define what was meant by "a suitable bumblebee" in this case. In this context, the performance of a bumblebee was considered "suitable" if it successfully navigated the path from the entrance of the arena to the feeder and back via the designated path within a time frame of five minutes or less. To verify this condition, two trials with the trained settings (p8.0, u0.0, train (no rotation)) were examined. In the event that the bumblebee successfully completed both trials within the designated five-minute timeframe, it proceeded with further experimentation.

The experimental design involved subjecting the bumblebees to varying levels of polarized light, with each bee undergoing testing in unique trials that incorporates distinct settings.

Table 1 - The different settings and conditions used in the experiment

P refers to polarized light, while u refers to unpolarized light

In "vertical" direction, the axis of the angle of polarization is aligned with the path from the feeder to the entrance, "horizontal" describes the axis of the angle of polarization, which was 90° from the axis.

experimental setting	stimulus rotation	stimulus direction	DOP	intensity of stimulus [$\mu\text{W}/\text{cm}^2$]
p0.0 u1.0	TB	vertical to horizontal	0.07	3
p0.7 u1.0	TB	vertical to horizontal	0.07	3
p1.3 u1.0	TB	vertical to horizontal	0.08	3
p1.5 u1.0	TB	vertical to horizontal	0.08	3
p1.8 u1.0	TB	vertical to horizontal	0.10	3
p2.5 u1.0	TB	vertical to horizontal	0.12	3
p4.4 u1.0	TB	vertical to horizontal	0.15	3
p8.0 u1.0	TB	vertical to horizontal	0.19	3

p0.4 u2.0	TB	vertical to horizontal	0.04	11
p0.9 u2.0	TB	vertical to horizontal	0.07	11
p1.5 u2.0	TB	vertical to horizontal	0.05	11
p8.0 u2.0	TB	vertical to horizontal	0.12	11
p0.0 u8.0	TB	vertical to horizontal	0.07	101
p3.8 u8.0	TB	vertical to horizontal	0.07	101
p4.3 u8.0	TB	vertical to horizontal	0.07	101
p7.7 u8.0	TB	vertical to horizontal	0.08	101
p8.0 u8.0	TB	vertical to horizontal	0.07	101
p0.7 u1.0	LR	horizontal to vertical	0.06	3
p2.0 u1.0	LR	horizontal to vertical	0.10	3
p2.5 u1.0	LR	horizontal to vertical	0.12	3
p3.5 u1.0	LR	horizontal to vertical	0.14	3
p4.2 u1.0	LR	horizontal to vertical	0.12	3
p4.4 u1.0	LR	horizontal to vertical	0.12	3
p8.0 u1.0	LR	horizontal to vertical	0.19	3
p0.0 u2.0	LR	horizontal to vertical	0.04	11
p0.6 u2.0	LR	horizontal to vertical	0.04	11
p1.5 u2.0	LR	horizontal to vertical	0.05	11
p0.1 u8.0	LR	horizontal to vertical	0.05	101

p3.8 u8.0	LR	horizontal to vertical	0.04	101
p4.2 u8.0	LR	horizontal to vertical	0.05	101
p4.4 u8.0	LR	horizontal to vertical	0.05	101
p8.0 u8.0	LR	horizontal to vertical	0.05	101
p8.0 u0.0	training	Vertical (no rotation)		

The user was able to adjust the different levels of intensity using the provided controls. The stimulus was rotated by turning the device by 90°, which led to a 90°-rotation of the angle of polarization.

3 ANALYSIS

The objective of this experiment was to investigate the behavior of path integration under different stimulus intensity of polarization. The ensuing discussion will address the following research questions. The study will examine whether the degree of polarization affects the bumblebee's ability to perform path integration during navigation. The following research question is proposed: do the bumblebees perform a more accurate home vector under stronger or under weaker polarization conditions? The interpretation of the term "accurate" refers to the level of scatter of how much the data is spread out. Furthermore, the study will explore whether bumblebees utilize their short-term memory when following a visual cue after its rotation, or whether they rely on their long-term memory.

A total of 77 trials from six different bumblebees with the identification 49w, 40w, 70g, 69w and 21w were recorded. A single trial was recorded for each bumblebee under a specific stimulus setting, with the exceptions of bumblebee 40w, p8.0, u0.0, training, for which two trials were recorded, bumblebee 69w underwent two trials for the setting p8.0, u1.0, TB and for p8.0 u0.0, training, while bumblebee 21w underwent three trials under the setting p8.0, u1.0, TB.

To investigate if the bumblebees used the provided polarization stimulus for their orientation and if the bumblebee's ability to orientate is affected by the degree of polarization, the data is analyzed. Additionally, an investigation was conducted to ascertain whether bumblebees employ an alternative mechanism for orientation in lower degrees of polarization. This circumstance results frequently from cloud cover leading to the absence of information regarding the skylight polarization pattern. Thus, it is essential to determine the angles from the feeder to the points at which the bumblebees exited the inner and outer circles. Therefore, the angles were measured by the program "ImageJ" (Image processing and Analysis in Java, <https://imagej.net/ij/>). The measured angles were documented in Table A, which is provided in the appendix. The emission of polarized light was limited to the inner circle. Consequently, the analysis is restricted to the data of the inner circle.

The results were calculated and visualized with the statistical programming language R version 4.3.1

The Rayleigh-test was used to identify if the distribution of the data of the measured angles was random or if there was a preferred direction.

Values for kappa and mu were calculated within the von mises distribution.

4 RESULTS

4.1 EFFECT OF DOP TO THE PATH INTEGRATION BEHAVIOR

In the left panels of figures 9-13, the design is similar to the setup of the arena as viewed from below. As the inner circle, and therefore the area, in which the polarization was emitted, was constructed to be circular, the circular plots are a suitable presentation of the results. Only inner circle data is analyzed, as in this area no polarization was emitted to use as a cue for orientation and the distraction by the visible curtain was observable. The cross in the center of the circle represents the feeder. Each dots on the edge of the circle symbolize the measured bearing (angle) of each trial, where the bumblebees left the inner circle. Their real nest entrance, and therefore their most direct home vector, is located at 180°.

The right plot shows the result of the Maximum Likelihood Estimation (Circ-MLE) (Fitak and Johnsen, 2017) and is therefore, a suggestion of a suitable model. The bars show the number of points in each sector plotted as a circular histogram, whereas the dashed oval shows the estimated probability of an exit at that angle. Circ-MLE tests if the data is uniform (no specific orientation), unimodal (only one main orientation) or bimodal (two main orientations). For each model, it calculates the Akaike Information Criterion (AIC) value (Akaike, 1974) and therefore it suggests a function of likelihood and penalized by model complexity. The smaller the AIC-value, the better the suggested model fits. The calculated AIC-values are provided in the appendix.

Training Trials

First, the data of the training trials were investigated, as the bumblebees were foraging under the same stimulus conditions as in the training phase (p8.0, u0.0, no rotation). Consequently, the angle of the polarization stimulus remains aligned with the axis between the feeder and the nest entrance during the trials and the expected direction of the bumblebees is likely aligned with the stimulus orientation.

In data from these trials, the measured bearings are not distributed equally (Figure 9). There are two areas, one clustered around 120° and the other one around 300°. There is no unimodal orientation which is proven with the Rayleigh-Test ($p > 0,05$ with $r = 0,15$ and $n = 7$). The Circ-MLE suggests a bimodal model (M3A), with $\mu_1 = 295,70^\circ$, $\kappa_1 = 4,06$ and $\mu_2 = 115,67^\circ$, $\kappa_2 = 4,06$. As the values of kappa are higher than 1, the data in the two clusters is highly concentrated, and therefore pointing to two certain directions.

The adapted Rayleigh-Test of 2 θ (Batschelet, 1981) for bimodal models was not significant ($p_\theta > 0,05$ with $r = 0,54$ and $n = 7$). Consequently, it is a weak statistical evidence of a bimodal distribution. Still, there is a qualitative tendency in the two plots of being a bimodal distribution. This bimodal behavior is also shown in Patel's study (Patel et al., 2022), and indicated the orientation with the use of the polarization stimulus.

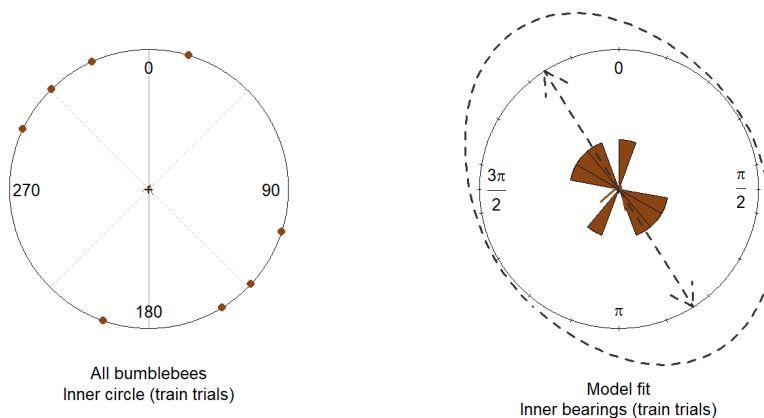


Figure 9 – Visualization of the raw data of the “training, inner circle” trails in a circular plot and representation of the bimodal Circ-MLE

The feeder is located at the center (cross). Each dot at the edge of the circle represents a measured bearing of the inner circle. In the right, a bimodal distribution is suggested. The bars show the density for the estimated distribution. The dashed oval is the confidence interval for the estimate.

Table 2 – The calculated values of different statistical tests and the adapted tests for a bimodal distribution for “training, inner circle” trials

Condition	p (Rayleigh)	r (test statistic)	n	p_θ	r_θ	n_θ
Train_InnerCircle	0,87	0,15	7	0,13	0,54	7

$\mu(\text{Mu}) [^\circ]$	κ (Kappa)	$\mu_1 [^\circ]$	κ_1	$\mu_2 [^\circ]$	κ_2	BestModel
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300,47	0,29	295.70	4.06	115.67	4.06	M3A
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As the representation of the data of all the different trials across stimulus rotation doesn't discriminate between strong and weak polarization (Appendix Figure A) and of its effect on the bumblebees ability is not detectable, it is not informative. Thus, strong and weak polarization have to be examines independently.

In the further analysis, the conditions p3.5 u1.0 ($DOP_{LR} = 0,14$), p4.2 u1.0 ($DOP_{LR} = 0,12$), p4.4 u1.0 ($DOP_{LR} = 0,12$; $DOP_{TB} = 0,15$), p8.0 u1.0 ($DOP_{LR, TB} = 0,19$), p8.0 u0.0 and p8.0 u2.0 ($DOP_{TB} = 0,12$) are referring to strong polarization. Weak polarization is considered as the conditions p3.8 u8.0 ($DOP_{LR} = 0,04$; $DOP_{TB} = 0,07$), p4.2 u8.0 ($DOP_{LR} = 0,05$), p4.3 u8.0 ($DOP_{TB} = 0,07$), p4.4 u8.0 ($DOP_{LR} = 0,05$), p7.7 u8.0 ($DOP_{TB} = 0,08$) and p8.0 u8.0 ($DOP_{LR} = 0,05$; $DOP_{TB} = 0,07$).

Strong polarization, LR

In the left-right condition, the orientation of the stimulus was rotated and therefore the axis of the angle of polarization was shifted 90° to the path from the feeder to the nest entrance. Thus, on the way back to the nest, the bumblebees experienced the same orientation of the stimulus as in the training phase. As the rotation of the stimulus during feeding could lead to a conflict by the bumblebees experiencing a change in the direction compared to their outbond path, it remains the same as in the familiar training condition. Additionally in this data set, the degree of polarization was high and thus, the stimulus provided strong information.

The data presents three different clusters of the bearings where the bumblebees left the inner circle (Figure 10). There is one around 0°, one is orientated to the opposite around 180° and the third one is around 260°. The short arrow in the plot averages all bearings and indicates the mean vector. The Circ-MLE suggests an uniform distribution with weak concentrated data ($M1$, $\mu = 207,33^\circ$, $\kappa = 0,76$) Additionally, the Rayleigh test is not significant ($p > 0,05$ with $r = 0,36$ and $n = 9$). As the AIC values for a uniform and a bimodal distribution are quite similar ($AIC_{M3A} = 24,76$ and $AIC_{M1} = 25,73$), a tendency towards a bimodal distribution around 0° and 180° can be assumed, which is also observable in the circular plot.

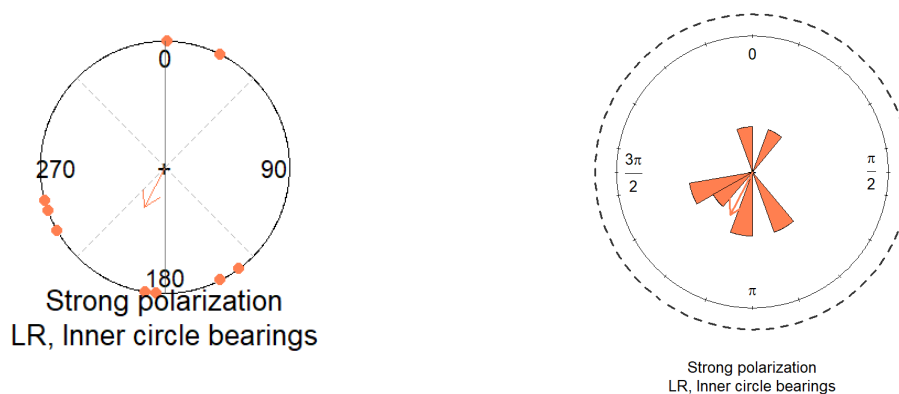


Figure 10 – Visualization of the raw data of the “strong polarization, LR, inner circle” trails in a circular plot and representation of the bimodal Circ-MLE
The arrow indicates the mean of the direction vector and is proportional to the mean vector length. A shorter arrow refers to a weaker concentration of the data.

Table 3 – The calculated values of different statistical tests for “strong polarization, LR, inner circle” trials

Condition	p (Rayleigh)	r (statistical test)	n	$\mu(\text{Mu}) [^\circ]$	κ (Kappa)	BestModel
StrongPol_LR	0,33	0,36	9	207,33	0,76	M1

Weak polarization, LR

In trials with weaker polarization, the bumblebees are expected to orient less accurately, but they are still able to react to the polarization stimulus.

As previously described, in LR conditions, the stimulus is in a vertical position after rotation, which is the direction the bumblebees should be familiar from training, but 90° to the orientation they observe while walking to the feeder.

The measured angles point to a higher concentration of data around 160° (Figure 11). This is also indicated by the mean vector, which is pointing to 160° . The arrow is slightly longer than in the representations of the data of the strong polarization conditions. Circ-MLE suggest an unimodal distribution (M2B). Still, as the Rayleigh test is not statistically significant ($p > 0,05$ with $r = 0,31$ and $n = 4$) and the value for kappa is < 1 ($\kappa = 0,67$), there is a weak concentration of the bearings.

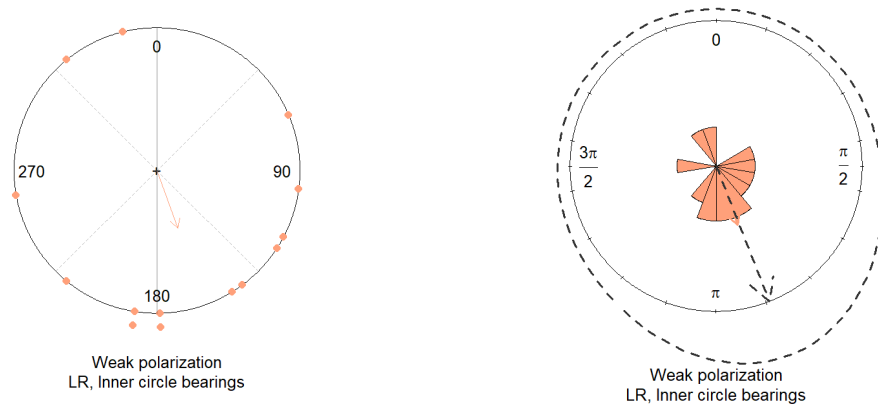


Figure 11 - Visualization of the raw data of the “weak polarization, LR, inner circle” trails in a circular plot and representation of the bimodal Circ-MLE

Table 4 - The calculated values of different statistical tests for “weak polarization, LR, inner circle” trials

Condition	p (Rayleigh)	r (statistical test)	n	$\mu(\text{Mu}) [^\circ]$	κ (Kappa)	BestModel
WeakPol_LR	0,70	0,31	4	19,15	0,67	M2B

Strong polarization, TB

For the top-bottom condition, the bumblebees were able to enter the arena in the same stimulus orientation as in training. After feeding at the conical feeder, the stimulus and therefore the angle of polarization was rotated from being aligned with the axis from the feeder to the nest entrance to being 90° shifted to this axis. If they were reacting to the stimulus, they were expected to exit the inner circle at angles around 90° and 270°, which are pointing to a fictive nest. If the bumblebees relied on other information, they were expected to walk towards the real nest.

The circular plot shows a wide distribution of the data. The most measured angles are occurring around 0°, which is also indicated by the arrow which points to the mean vector direction close to 0°. As the Rayleigh test is not statistically significant ($p > 0,05$ with $r = 0,15$ and $n = 15$) and the length of the mean vector is short, the data is weakly concentrated ($\kappa = 0,31$). An uniform model (M1) fits this distribution the best, as the data doesn't show a preferred direction.

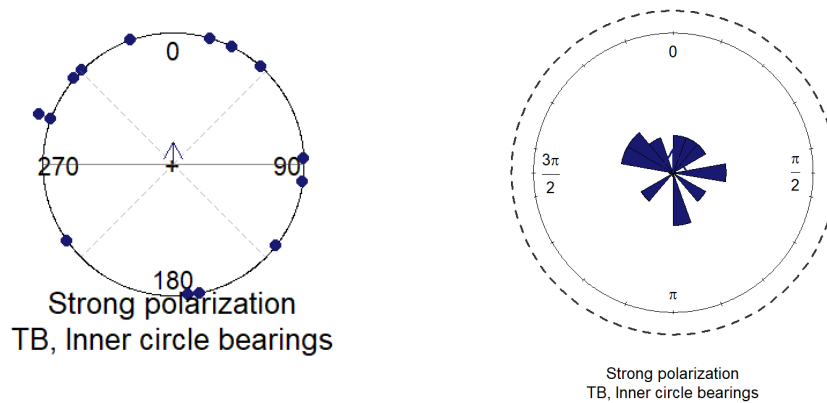


Figure 12 - Visualization of the raw data of the “strong polarization, TB, inner circle” trails in a circular plot and representation of the bimodal Circ-MLE

Table 5 - The calculated values of different statistical tests for “strong polarization, TB, inner circle” trials

Condition	p (Rayleigh)	r (statistical test)	n	$\mu(\text{Mu}) [^\circ]$	κ (Kappa)	BestModel
StrongPol_TB	0,71	0,15	15	24,78	0,31	M1

Weak polarization, TB

If the bumblebees are using the provided stimulus for their path integration behavior to return back to their nest after walking to a food source, the bumblebees are expected to walk to a fictive home in horizontal direction, as it is again the top-bottom condition.

In the circular plot, the measured bearings, presented as dots, are stronger clustered in the lower part than in the upper part of the plot. The calculated mean vector points to 240° ($\mu=241,79^\circ$). Although the Rayleigh test is not statistical significant ($p>0,05$ with $r = 0,45$ and $n = 6$), Circ-MLE suggest a unimodal distribution model (M2B). As the value of kappa is around 1 ($\kappa = 1,01$), there is a higher concentration of the measured bearings, compared to the conditions described before.



Figure 13 - Visualization of the raw data of the “weak polarization, TB, inner circle” trails in a circular plot and representation of the bimodal Circ-MLE

Table 6 - The calculated values of different statistical tests for “weak polarization, TB, inner circle” trials

Condition	p (Rayleigh)	r (statistical test)	n	$\mu(\text{Mu}) [^\circ]$	κ (Kappa)	BestModel
WeakPol_TB	0,31	0,45	6	241,79	1,01	M2B

For better understanding of whether the bumblebee`s chosen path towards its true nest or to the fictive nest direction as indicated by the angle of polarization, data were separated into quadrants around the true and fictive nest (Figure 14-15).

In LR condition, the fictive nest correlates with their real nest, as the angle of polarization during their path back to the nest is the same as experienced during training. In both DOP conditions, strong and weak, the bumblebees tend to walk back to the fictive nest (Figure 14), or in this case, to their real nest. To compare the result statistically and to prove if the probability of orientation is higher than 0,5 , a binomial test was performed. In strong condition, 6 of 9 (percentage of 0,67) individuals walked to the fictive nest. There result of the test shows no significant difference ($p= 0,254$; 95%-CI: [0,34; 1,00]). In weak condition, 9 of 14 (percentage of 0,64) bumblebees chose to walk back to the fictive nest. Still, the test was not significant ($p = 0,21$; 95%-CI: [0,39; 1,00]). Even if the test don`t show significant results, a tendency towards the fictive nest, where the angle of polarization is aligned with the same axis as in training,

is observable. In weaker conditions, there is stronger orientation to the fictive nest compared to the stronger conditions.

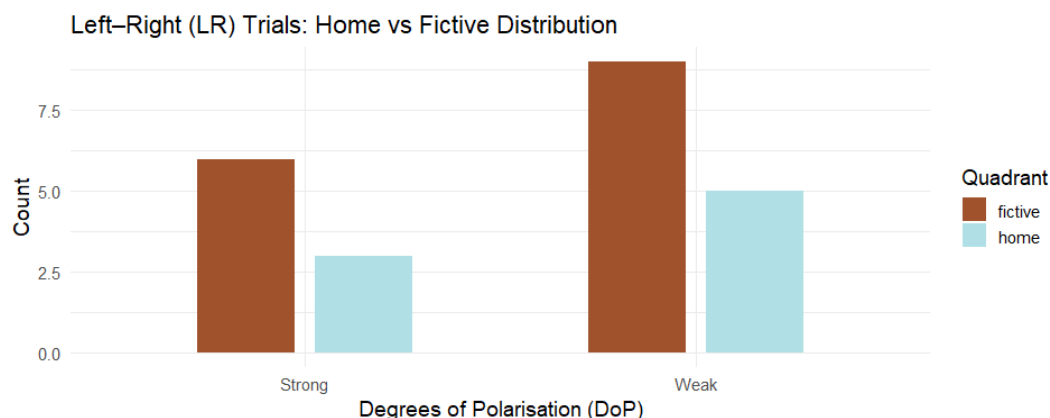


Figure 14 – The amount of chosen paths either to their home or to their fictive nest in LR condition dependent on the degree of polarization

The x-axis represents whether it was the conditions with stronger polarization or the ones with weaker conditions. On the y-axis, the number of path decisions to the fictive nest, which is designed in brown color or their home in blue is shown. In LR condition, the fictive nest (brown) correlates with their real home (blue), as the direction of the polarization points to their nest because of its horizontal setting. In weaker and in stronger polarization settings the bumblebees tend to walk to the fictive (and therefore their real home). In weaker conditions, this behavior is even more frequent

Whereas the distribution of the path back to the fictive nest or their home is equal in the TB condition (Figure 15) with stronger polarization settings. 7 of 14 (percentage of 0,50) bumblebees orientated towards the fictive nest, which shows no significance in the binomial test ($p = 0,61$; 95%-CI: [0,26;1,00]). In weaker polarization settings, 7 of 12 (percentage of 0,58) bumblebees tend to walk to the fictive nest. The binomial test was not significant ($p = 0,39$; 95%-CI: [0,32; 1,00]). In TB condition, the fictive nest and their real home are 90° shifted.

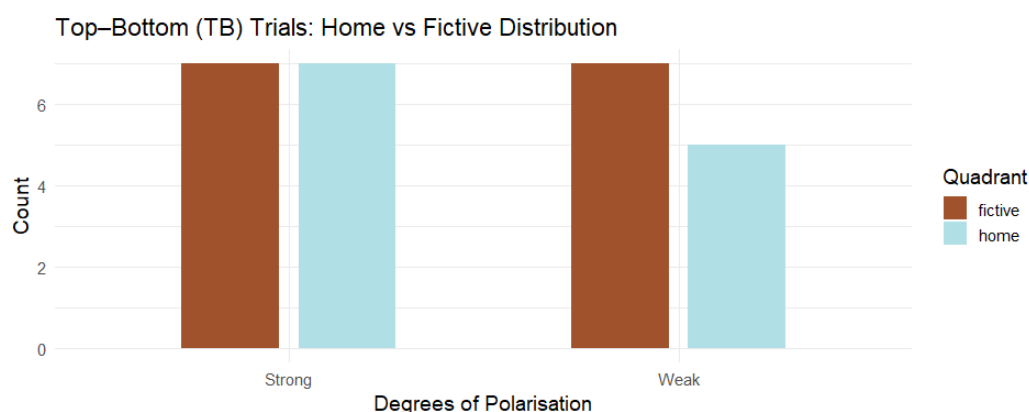


Figure 15 - The amount of chosen paths either to their home or to their fictive nest in LR condition dependent on the degree of polarization

In TB condition, their fictive nest (brown) is 90° shifted to their real home (blue). In stronger polarization settings, the chosen way back is equal between the fictive and their real home, whereas in weaker polarization, the count of bumblebees walking back to the fictive home is slightly higher.

4.2 MOST ACCURATE HOME VECTOR

To further analyze, whether there was a condition, in which the bumblebees were performing a more accurate home vector than in the other polarization condition, it is necessary to calculate the deviation from the real home angle to every other measured angle from each trail. The real nest is thereby located at 180°. To verify the results statistically, a Rayleigh test was conducted. The Rayleigh test proves if the population of the data has a certain tendency or if it is random. The results are presented in a Boxplot (Figure 16).

In the circular plot (Figure 16), the measured bearings are distributed differently in all the different conditions. It is also observable that the amount of measured bearings differs significantly in all the four conditions. A visual tendency towards the nest is observable in the strong TB condition.

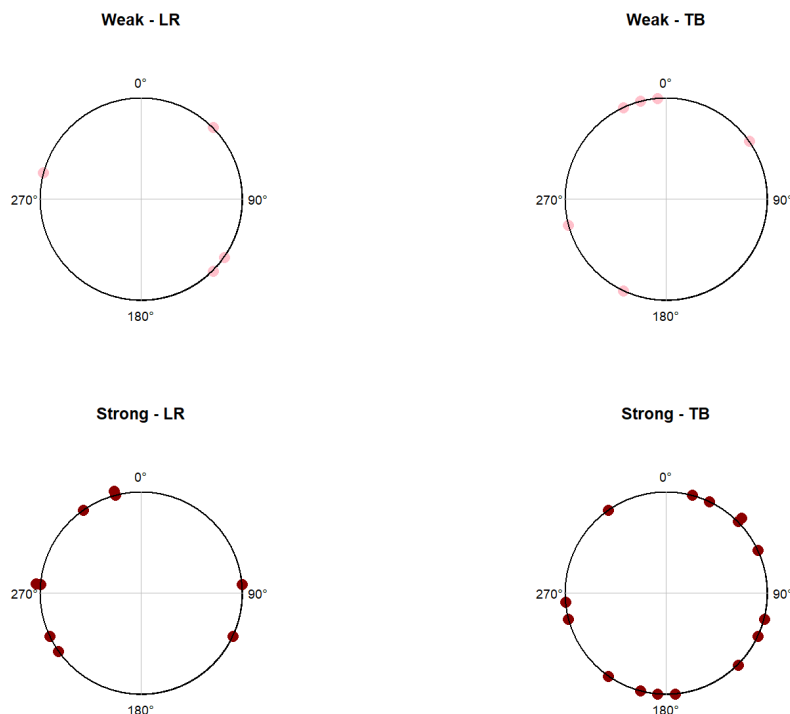


Figure 16 – Circular plots of the stronger and weaker polarization for LR and TB condition

The weaker condition is indicated as pink dots, whereas the dark red dots present the stronger polarization. A distinction is made between the LR and TB trials

Consequently, the data is presented in a bar plot (Figure 16) and in a Boxplot (Figure 17), to examine the data even better.

The bar plot of the LR condition shows, that the mean deviation is lower in stronger polarization conditions (about 70°) than in weaker conditions (about 110°). In TB conditions, the mean deviation is higher in stronger polarization conditions (about 100°) compared to the weaker condition (about 75°).

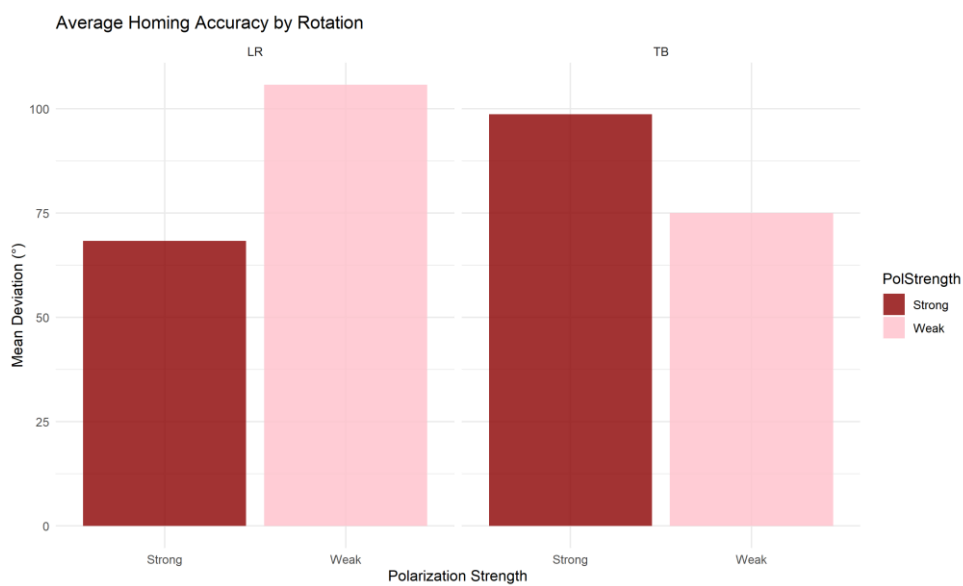


Figure 17 - Bar plot of the average homing accuracy of strong (dark red) and weak (pink) polarization settings in LR and TB condition

On the x-axis, the polarization strength, categorized as LR and TB, of LR and TB condition, is shown. The y-axis shows the mean deviation in degrees. The lower the value of the mean deviation, the more accurate is the orientation towards the nest. The bars are presenting the mean deviation in degrees.

In LR condition, the mean deviation of the stronger polarization condition is lower, whereas the weaker polarization condition in TB trials leads to lower deviation.

To examine this data even better, a boxplot was used for further investigation (Figure 17). The boxplot shows the deviation from the nest. The calculated median for the stronger LR condition is lower (around 55°) compared to the weaker LR condition (around 125°). The interquartile range of weaker LR conditions (around 40°) exhibits a slight narrowing compared to the range of stronger LR conditions (around 50°). However, an upward spatial shift is observed in the graph. For TB trials, the median of the stronger polarization conditions is higher (around 110°) than the median of the weaker polarization conditions (around 75°). There is a higher dispersion of the

interquartile range in stronger conditions (around 80°) compared to the weaker polarization conditions (around 20°).

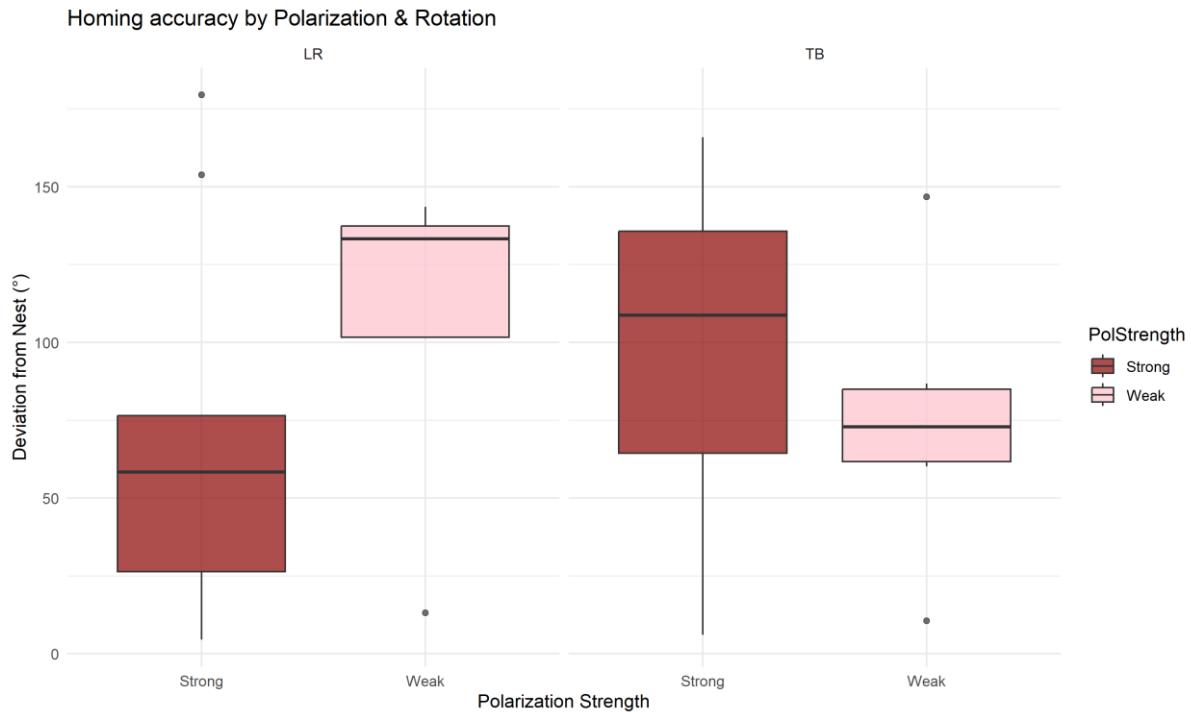


Figure 18- Boxplot of the Deviation from the Nest in weak and in strong polarization conditions for LR and TB condition
For the LR condition, the interquartile range of the strong polarization condition (dark red) is around 50° with a median around 55°, whereas the interquartile range in weaker conditions (pink) is about 40° with a median of 125°. For the TB condition, the interquartile range of the strong polarization condition is around 80° with a median around 110° and the interquartile range in weaker conditions is about 20° with a median of 75°

The calculated mean deviations of the stronger LR and the weaker TB conditions are quite similar ($MD_{\text{Strong,LR}} = 68,36^\circ$; $MD_{\text{Weak,TB}} = 75,03^\circ$). Additionally, the mean deviations of the LR trials in weaker polarization and the trials of TB condition in stronger polarization are around 100° ($MD_{\text{Weak,LR}} = 105,79^\circ$; $MD_{\text{Strong,TB}} = 98,71^\circ$). The standard deviation is the lowest in weaker TB condition ($SD_{\text{Weak,TB}} = 44,20^\circ$), slightly higher in strong TB condition ($SD_{\text{Strong,TB}} = 50,78$) and around 62° for both LR conditions ($SD_{\text{Weak,LR}} = 61,96^\circ$; $SD_{\text{Strong,LR}} = 61,66^\circ$)

The Rayleigh test is in strong and weak polarization for LR and TB condition not significant ($p > 0.05$), thus there is no specific orientation in all conditions.

Table 7 – The results of the mean deviation (MD), the standard deviation (SD) and the statistical Rayleigh test of strong and weak polarization in LR und TB conditions

Stimulus rotation	Polarization	n	MD[°]	SD [°]	Rayleigh p	Rayleigh r
LR	Strong	9	68,36	61,66	0,33	0,36
LR	Weak	4	105,79	61,96	0,70	0,32
TB	Strong	15	98,71	50,78	0,71	0,15
TB	Weak	6	75,03	44,20	0,31	0,45

4.3 USAGE OF SHORT- OR LONG-TERM MEMORY

In the event that the bumblebees are following the stimulus after its rotation, which either leads to a 90° shift of their home path in TB condition or to a conflicting direction change, in LR conditions, the bumblebees are using their short-term memory (STM). In LR condition, it was a conflicting direction change, as the rotation indicated a change in their orientation after feeding, but the angle of polarization was the same axis as in training after rotation. If the bumblebees don't respond to the stimulus, they are walking back to their real nest independently of the provided stimulus. In this case, the bumblebees are comparing their memory of the stimulus from training to the current stimulus and therefore are using their long-term memory (LTM). To investigate this behavior, first the areas of the angles for short- and long-term memory have to be defined.

If the bumblebees are using their short-term memory, the sector from 45° to 135° and the one from 225° to 315° are referred to, which is shown with the light blue coloring. For the usage of the long-term memory, the angles within the sector from 315° to 45° and the sector from 135° to 225° in light red coloring are considered. This division was carried out, because in TB condition, the bumblebees should walked to the short-term memory area if reacting to the stimulus, as the angle of polarization is aligned with an axis 90° shifted to the feeder and nest entrance axis. In LR condition, the bumblebees experienced a conflict situation, as they experienced a rotation of the angle of polarization during feeding, but it was then aligned with the axis from training. If the bumblebees were reacting to the rotation, their path was 90° shifted and therefore in the short-term memory sector. If they were combining their memory from the current

situation and from training, as the angle of orientation was the same as in training, this would have led to an orientation towards the long-term memory sector. Those sectors are presented in Figure 19.

Short and long term memory Regions

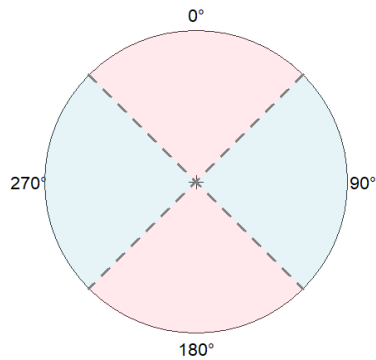


Figure 19 – Assignment of the bumblebee's orientation with the usage of their short- or long-term memory
In light red, the region of the long-term is indicated (313°-45°, 135°-225°). If the bumblebees are using their short-term memory for orientation, the measured bearings are within the light blue area (45°-135°, 225°-315°)

To get clear results of the usage of long- or short-term memory, the distribution in percentage by condition is shown in the bar chart (Figure 20). For the LR condition, 66,7% of the measured bearings are located in the long-term memory region, while 33,3% are positioned in short-term memory areas.

In TB condition, the percentage of the bearings within the short-term regions are quite higher (63,6%). 36,4% of the bearings of TB condition are placed within the long-term memory area. The Chi² test suggest a significant dependency between LR or TB condition and the use of the long-term or short-term memory ($p < 0,05$; $p = 0,04$).

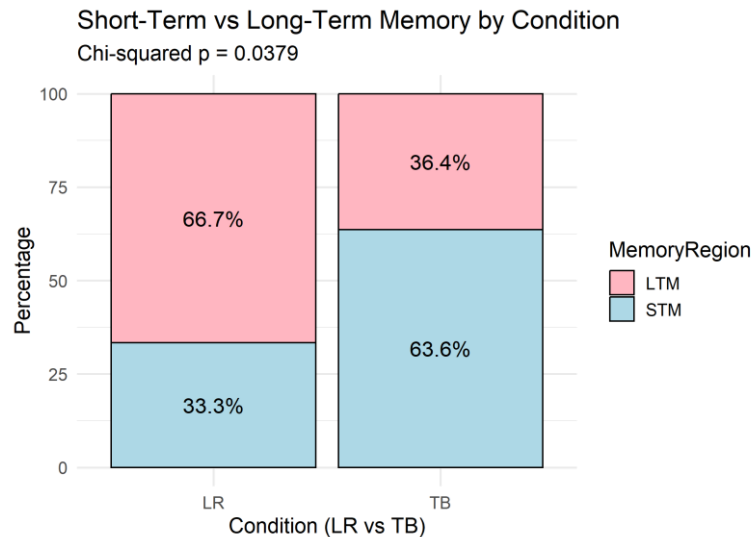


Figure 20 – Distribution of the long- (LTM) or short-term memory (STM) in percentage for LR and TB condition
In LR condition, 66,7% of the bearings are placed within the LTM area and 33,3% are within the STM area. For TB condition, more bearings are located around the area of the STM (63,6%) , whereas 36,4% are within the LTM regions. The χ^2 -Test has a value of $p=0,0379$

Looking at all the measured bearings (Figure 21), distinguished between LR condition (blue dots) and TB condition (red dots), the majority of the measured angles of the TB conditions are located in the STM area, whereas the angles in LR condition are more frequently located in the LTM region.

Memory Regions: STM vs LTM

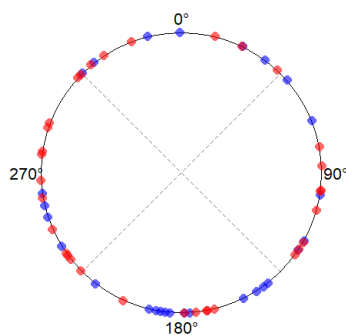


Figure 21 – Representation of the measured bearings, distinguished between LR (blue) and TB (red) in the different region for short-term memory (STM) and long-term memory LTM)
More blue dots are visible in the LTM region, indicating that the use of the long-term memory was more present in the LR condition. In the short-term memory areas, there is a higher amount of red dots for TB condition.

5 DISCUSSION

5.1 EFFECT OF DOP TO THE PATH INTEGRATION BEHAVIOR

Some studies (Guan et al., 2018; Homberg et al., 2011; Horváth, 2014) have already discovered that bumblebees use the polarization patterns of the sky for navigation. In this experiment, the bumblebees are therefore expected to use the provided artificial polarization stimulus as a cue providing information for their home path and therefore, to be essential for their foraging path integration behavior. The question remains whether the degree of polarization affects on the path integration behavior of bumblebees (Edrich and Helversen, 1987; Foster et al., 2019; Henze and Labhart, 2007).

In the training trials, the polarization direction was designated to be aligned with the axis of the feeder to the nest entrance, equal to the setting as in the training phase. Thus, the expected home path is either one pointing to their nest entrance around 180° or, as the provided polarization stimulus is a bimodal stimulus (Foster, 2024; Heinze et al., 2018), pointing to the opposite side, the direction towards their nest entrance around 0° .

The measured bearings of the experimental data in the training trails are either concentrated close to 300° or to 120° ($\mu_1=295,7^\circ$, $\kappa_1 = 4,06$ and $\mu_2=115,67^\circ$, $\kappa_2 = 4,06$). As their real nest entrance is located at 180° , there is a deviation of their home path, which does not point straight back to their nest entrance. Still, as a definite bimodal behavior is observable, it can be concluded that the bumblebees are using the provided stimulus to orient themselves, as some bearings were observed to the opposite of the nest entrance. The angle of polarization is aligned with the axis from the feeder to the opposite. As the bumblebees walked from the nest entrance to the feeder, they could learn to go this path back and forward. If they are going to the opposite direction, they have not walked this path before. Therefore, they rely on the provided polarization stimulus to orient.

To analyze if the degree of polarization has an effect to the path integration behavior of the bumblebees, it was necessary to distinguish between strong and weak polarization to further observe the differences in behavior between a strong and a weak provided stimulus. With weaker polarization, the bumblebees are expected to become

less accurate in their orientation, as they are not able to detect the polarization cue with their dorsal rim area and therefore don't receive enough information about their location to use for their path integration. With higher DOP, the bumblebees are predicted to perform better than in the weak condition.

In case that the bumblebees are using the provided stimulus for their orientation, for the TB condition, it is anticipated that the bumblebee walks to a 90° shifted horizontal direction towards the nest and therefore to the angles 90° or the opposite 270° after feeding at the feeder and the simultaneous rotation of the polarization. In LR condition, the polarization information is aligned with the vertical direction of their home. But it is considered as a conflicting stimulus, as the orientation of the stimulus after rotation is the same as in the training, but the bumblebees experience a change of the stimulus during foraging. Thus, the bumblebees could walk a 90° shifted direction towards the nest, if following the stimulus rotation. If they are following the angle of polarization, they are expected to walk back to the nest, as the axis of the angle of polarization is the same as in training conditions after rotation.

Looking at the LR condition data, there is a tendency of the bumblebees to walk close to the nest or to the opposite direction ($\mu_{\text{strong, LR}} = 207,33^\circ$, $K_{\text{strong, LR}} = 0,74$ and $\mu_{\text{weak, LR}} = 19,15^\circ$, $K_{\text{weak, LR}} = 0,67$). This would lead to the assumption that the bumblebees still orient toward the trained home direction, even under conflicting condition. This could support the hypothesis of the use of long-term memory, which will be discussed further. Comparing the strong and weak polarization of the LR condition, there is no clear decreasing trend of the bumblebee's ability to find back home in stronger polarization.

If the TB trials with weak and strong polarization settings are compared, the bumblebees seem to perform better in weaker polarization. As the measured bearings of the strong polarization, TB are widely spread and the suggested model was an uniform distribution with weak concentrated data ($K_{\text{strong, TB}} = 0,31$), there is no preferred direction visible. In trials with weaker polarization, the preferred stronger concentrated direction is around the angle of 240° ($\mu_{\text{weak, TB}} = 241,79^\circ$, $K_{\text{weak, TB}} = 1,01$). Still, although it is the TB condition and the bumblebees are expected to walk back home to a fictive home 90° shifted from the real home, the measured bearings don't support this assumption as their orientation is more likely to be aligned with their real home. This might support the hypothesis of the additional use of the long-term memory. The

bumblebees learned the way back from previous foraging trips and saved the information to use it for navigation, too.

Additionally, the Rayleigh test is not statistically significant ($p > 0,05$) in all the different observed conditions, which leads to the conclusion that there is no evidence for a preferred direction and therefore the orientations of the bumblebees are statistically random. Still, the low amount of samples from different investigated trials should be in mind.

Furthermore, the direction of the provided polarization stimulus is important, as it is shown in Figure 13 and Figure 14. In the LR condition, the fictive nest, indicated by the vertical direction of the stimulus, correlates with their home. Thus, the bumblebees exhibit a higher tendency to go back to the fictive nest. As it is the same case in weaker condition, the bumblebees are not necessary able to use the provided stimulus and therefore the use of another potential information for orientation could be possible. The bumblebees could receive this information through the usage of their long-term memory.

To summarize, the bumblebees still orient toward the home direction in the TB and even under the conflicting LR condition. In some trials, the bumblebees are reacting to the stimulus, but some of the bumblebees doesn't seem to react to the direction of the polarization at all. Although the data of the training trails indicated that the bumblebees are reacting to the stimulus, a higher concentrated clustering of the bearings closer to 0° and 180° were expected. With higher degree of polarization, it was predicted that the bumblebees were strongly reacting to the stimulus and moved towards the true or fictive nest, as a higher degree of polarization offers a stronger and more reliable cue. Other than expected, the bumblebees orientation towards their real home seems to be better in weaker TB conditions.

This result could suggest that the bumblebees are not performing their path integration with the only use of polarization. This behavior would be important, if the skylight polarization patterns would not be available or less available in the bumblebee's natural environment as it would be the case in cloudy or foggy conditions. Furthermore, the bumblebees could learn to find their way back to their home with using their knowledge to walk back, which indicates the usage of their long-term memory. Either way, as the Rayleigh test was not significant in all of the observed conditions, a larger sample of the different trials is obligatory to get clear evidence.

5.2 MOST ACCURATE HOME VECTOR

For the analysis of the question, if a more accurate home vector occurs in weaker or in stronger polarization conditions, a bar plot and a Boxplot examined angular deviations from the expected direction.

The bar plot of the two LR and TB conditions show, that in stronger LR condition, the orientation towards the nest is more accurate, whereas the bumblebees orientate more accurate towards the nest in weaker TB condition.

The use of a box plot further enhances the clarity of this outcome. In the stronger LR conditions, the median is lower (around 55°) compared to the weaker condition (around 125°). For TB trials, the median is lower (around 75°) and the interquartile range is narrower in weaker condition (around 20°) compared to the stronger polarization condition. As the data shows no uniform result with regard to the accuracy for a certain degree of polarization. The bumblebees walk more accurate towards the nest in strong polarization condition in LR condition, but they orientate less accurate in strong polarization in TB trials. This result suggests, that the combination of angle of polarization and rotation direction have an effect to the accuracy. Additionally, the bumblebees could rely on their long-term memory to navigate back to their home.

The calculated mean deviation of the stronger LR and the weaker TB conditions ($MD_{strong,LR} = 68,36^\circ$; $MD_{weak,TB} = 75,03^\circ$) are lower than of the weaker LR and stronger TB conditions ($MD_{weak,LR} = 105,79^\circ$; $MD_{strong,TB} = 98,71^\circ$). The standard deviation is the lowest in weaker TB condition ($SD_{weak,TB} = 44,20^\circ$) and stronger TB condition ($SD_{strong,TB} = 50,78$) and around 62° for both LR conditions ($SD_{weak,LR} = 61,96^\circ$; $SD_{strong,LR} = 61,66^\circ$). These results further enhances the assumption of the importance of the usage of their long-term memory. The hypothesis that the bumblebees are less prone to not finding their way to the nest when clouds appear is predicated on the assumption that they have the capacity to fall back on their long-term memory.

Still, no significant deviation in both polarization conditions of LR and TB trials were detectable in the Rayleigh test ($p > 0,05$). Thus, there is no statistically specific orientation in all conditions.

Consequently, there is no evidence that the bumblebees perform a better home path vector with a higher accuracy towards the nest for any degrees of polarization. Instead,

the combination of angle of polarization and rotation is essential for their accurate orientation.

Additionally, instead of the polarization stimulus, the bumblebees could use different information to orient and to construct a home path vector. The flexible use of a long-term memory is suggested. To further analyze this, the third research question must be answered.

For better results, more trials for one polarization setting per one bumblebee are required, as the different bearings for one setting would then be able to compare.

5.3 USAGE OF SHORT- OR LONG-TERM MEMORY

In the LR condition, the majority (66,7%) of the bumblebees were walking back to a direction within the long-term memory area. In the TB condition, more bumblebees walked a home path vector pointing to a direction within the short-term memory region (63,6%). The usage of the certain memory type is therefore significantly dependent on the rotation condition ($p < 0,05$).

This result also correlates with the provided direction of the polarization stimulus. In TB condition, the stimulus is 90° shifted to their real home, therefore horizontal and the indicated direction of the stimulus is pointing towards the region of the short-term region. In LR condition, the stimulus is pointing vertical to the nest after rotation, indicating an orientation towards the long-term memory region.

Still in both cases, around 33% of the bumblebees were walking to the opposite memory area. Thus, in LR condition, 33,3% of the bumblebees used their short-term memory, while 36,4% of the measured angles in TB condition are located within the long-term memory area.

Consequently, the bumblebees are using both their short- and long-term memory dependent on the type of change in their visual environment. Their usage of the memory types is flexible. Thus, both memory types are obligatory for the bumblebee's ability to orient themselves and to find back to their nest.

6 CONCLUSION

The importance of skylight polarization patterns, caused by scattering of the sunlight, for the path integration behavior of some animals was already proven. The aim of this experiment was to investigate the effect of different intensities of visual artificial polarization patterns to the bumblebee's behavior performing a home path vector. The findings of the experiment, as well as the data evaluation, indicated that the impact of varying DOP values on the navigation behavior of bumblebees holds greater significance, instead of different intensities. Additionally, it was questioned if the degree of polarization affects the accuracy of their home vector. Furthermore, the usage of the bumblebee's short- and long-term memory dependent on the provided polarization stimulus was examined.

The bumblebees seems to react to the provided stimulus. Unexpectedly, a higher degree of polarization did not lead to a better orientation to the bumblebee's real home. After comparison of all the different trials with the different stimulus settings, the bumblebees seemed to perform better in weaker TB conditions. Still, in the most cases in weak and strong LR and TB conditions, the bumblebees are more likely to walk to a direction close to their home. This indicates the usage of other information.

The results of the question if the degree of polarization has an effect to the accuracy of the home vector don't show any evidence for this assumption. The ability of the bumblebees to walk back to their nest does not increase or decrease within stronger or weaker polarization. Instead, it is more dependent on the angle of polarization and its rotation change. Furthermore, the bumblebee may learn the certain path to their nest in their foraging trips and are able to use this learned information.

This assumption would lead to the question if the bumblebees are using their short- or their long-term memory to find back to their home. The results suggest the usage of both memory types depending on the provided cues or of the type of change in their environment. Thus, both, short- and long-term memory, are important for their survival and the combination of both memory types is essential for their orientation. Thus, the usage of a certain memory type is considered to be flexible.

Nonetheless, to get clear evidence of these data and for further analysis, the collecting of more data is obligatory. Furthermore, more positive (polarized 8,0; unpolarized 0,0)

and negative control trials (polarized 0,0; unpolarized 8,0) have to be collected. In this data set, the bumblebees were only investigated in positive control trials.

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APPENDIX

Table A – The measured angles of the different trials of the bumblebees 49w, 40w, 70g, 69w and 21w with different polarization settings and the two rotation conditions

Only one trial per one setting for one bumblebee was recorded. If there were exceptions, the trials were numbered from one to two or to three. The measured angles were calculated through the ImageJ software, and then transformed into bearings by computing $-\text{angle} + 90^\circ$ with R.

with TB: top bottom condition (rotation from vertical to horizontal), LR: left right condition (rotation from horizontal to vertical), i: inner circle, o: outer circle

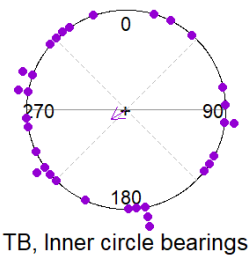
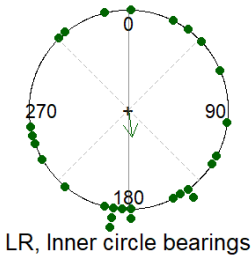
individual ID	intensity of polarisation	stimulus rotation	trial number	location of measured angle	Measured angle
49w	p0.0 u1.0	TB	1	i	-135.651
49w	p0.7 u1.0	TB	1	i	-141.818
49w	p4.4 u1.0	TB	1	i	135.716
49w	p8.0 u1.0	TB	1	i	-83.967
49w	p0.4 u2.0	TB	1	i	-7.184
49w	p0.9 u2.0	TB	1	i	129.864
49w	p1.5 u2.0	TB	1	i	-79.261
49w	p8.0 u2.0	TB	1	i	75.851
49w	p0.0 u8.0	TB	1	i	-88.568
49w	p3.8 u8.0	TB	1	i	123.179
49w	p4.3 u8.0	TB	1	i	-29.859
49w	p7.7 u8.0	TB	1	i	-79.408
49w	p8.0 u8.0	TB	1	i	-156.171
49w	p0.7 u1.0	LR	1	i	128.246
49w	p2.0 u1.0	LR	1	i	-127.488
49w	p3.5 u1.0	LR	1	i	90.444
49w	p4.4 u1.0	LR	1	i	-166.504
49w	p0.0 u2.0	LR	1	i	103.761
49w	p0.6 u2.0	LR	1	i	-33.563
49w	p1.5 u2.0	LR	1	i	-57.704
49w	p0.1 u8.0	LR	1	i	21.718
49w	p3.8 u8.0	LR	1	i	-103.163
49w	p4.2 u8.0	LR	1	i	53.531
49w	p4.4 u8.0	LR	1	i	41.139
49w	p8.0 u8.0	LR	1	i	134.690
49w	p8.0 u0.0	training	1	i	135.000
49w	p0.0 u1.0	TB	1	o	-90.991
49w	p0.7 u1.0	TB	1	o	-116.477
49w	p4.4 u1.0	TB	1	o	156.188
49w	p8.0 u1.0	TB	1	o	-89.802
49w	p0.4 u2.0	TB	1	o	-73.404
49w	p0.9 u2.0	TB	1	o	-93.202
49w	p1.5 u2.0	TB	1	o	-97.581
49w	p8.0 u2.0	TB	1	o	118.661

49w	p0.0 u8.0	TB	1	o	-124.906
49w	p3.8 u8.0	TB	1	o	20.266
49w	p4.3 u8.0	TB	1	o	-102.551
49w	p7.7 u8.0	TB	1	o	3.643
49w	p8.0 u8.0	TB	1	o	-175.304
49w	p0.7 u1.0	LR	1	o	-90.843
49w	p2.0 u1.0	LR	1	o	-92.053
49w	p3.5 u1.0	LR	1	o	-91.181
49w	p4.4 u1.0	LR	1	o	-129.806
49w	p0.0 u2.0	LR	1	o	-94.086
49w	p0.6 u2.0	LR	1	o	-33.576
49w	p1.5 u2.0	LR	1	o	-90.000
49w	p0.1 u8.0	LR	1	o	42.923
49w	p3.8 u8.0	LR	1	o	-91.023
49w	p4.2 u8.0	LR	1	o	28.802
49w	p4.4 u8.0	LR	1	o	-91.569
49w	p8.0 u8.0	LR	1	o	-1.073
49w	p8.0 u0.0	training	1	o	133.112
40w	p0.7 u1.0	TB	1	i	-33.079
40w	p1.5 u1.0	TB	1	i	-144.851
40w	p2.5 u1.0	TB	1	i	171.289
40w	p8.0 u1.0	TB	1	i	161.274
40w	p0.4 u2.0	TB	1	i	-114.370
40w	p8.0 u2.0	TB	1	i	-76.430
40w	p4.3 u8.0	TB	1	i	-169.611
40w	p7.7 u8.0	TB	1	i	-176.820
40w	p8.0 u0.0	TB	1	i	-15.479
40w	p0.7 u1.0	LR	1	i	-28.926
40w	p2.5 u1.0	LR	1	i	-171.357
40w	p4.4 u1.0	LR	1	i	-148.406
40w	p8.0 u1.0	LR	1	i	-161.417
40w	p0.0 u2.0	LR	1	i	-54.246
40w	p0.6 u2.0	LR	1	i	-98.443
40w	p1.5 u2.0	LR	1	i	-100.305
40w	p0.1 u8.0	LR	1	i	-88.798
40w	p8.0 u0.0	training	2	i	155.813
40w	p8.0 u0.0	training	1	i	-59.036
40w	p0.7 u1.0	TB	1	o	-118.874
40w	p1.5 u1.0	TB	1	o	51.877
40w	p2.5 u1.0	TB	1	o	-152.407
40w	p8.0 u1.0	TB	1	o	-90.000
40w	p0.4 u2.0	TB	1	o	-96.537
40w	p8.0 u2.0	TB	1	o	-122.784
40w	p4.3 u8.0	TB	1	o	-157.699
40w	p7.7 u8.0	TB	1	o	-171.783
40w	p8.0 u0.0	TB	1	o	-17.222
40w	p0.7 u1.0	LR	1	o	-119.843

40w	p2.5 u1.0	LR	1	o	95.899
40w	p4.4 u1.0	LR	1	o	-148.406
40w	p8.0 u1.0	LR	1	o	-114.562
40w	p0.0 u2.0	LR	1	o	-88.441
40w	p0.6 u2.0	LR	1	o	144.782
40w	p1.5 u2.0	LR	1	o	-122.177
40w	p0.1 u8.0	LR	1	o	-88.798
40w	p8.0 u0.0	training	2	o	143.043
40w	p8.0 u0.0	training	1	o	90.405
70g	p1.3 u1.0	TB	1	i	10.508
70g	p1.8 u1.0	TB	1	i	172.372
70g	p4.4 u1.0	TB	1	i	64.274
70g	p8.0 u1.0	TB	1	i	110.772
70g	p2.5 u1.0	LR	1	i	-9.071
70g	p3.5 u1.0	LR	1	i	-52.001
70g	p4.4 u1.0	LR	1	i	63.853
70g	p8.0 u0.0	training	1	i	74.529
70g	p1.3 u1.0	TB	1	o	-125.059
70g	p1.8 u1.0	TB	1	o	-121.649
70g	p4.4 u1.0	TB	1	o	-126.951
70g	p8.0 u1.0	TB	1	o	-161.432
70g	p2.5 u1.0	LR	1	o	-175.359
70g	p3.5 u1.0	LR	1	o	-89.799
70g	p4.4 u1.0	LR	1	o	87.010
70g	p8.0 u0.0	training	1	o	-90.199
69w	p4.4 u1.0	TB	1	i	47.021
69w	p8.0 u1.0	TB	1	i	2.816
69w	p8.0 u1.0	TB	2	i	137.317
69w	p2.0 u1.0	LR	1	i	-86.367
69w	p3.5 u1.0	LR	1	i	-96.600
69w	p8.0 u0.0	training	1	i	113.629
69w	p8.0 u0.0	training	2	i	-43.683
69w	p4.4 u1.0	TB	1	o	-177.831
69w	p8.0 u1.0	TB	1	o	1.691
69w	p8.0 u1.0	TB	2	o	-90.974
69w	p2.0 u1.0	LR	1	o	-90.774
69w	p3.5 u1.0	LR	1	o	-90.405
69w	p8.0 u0.0	training	1	o	102.048
69w	p8.0 u0.0	training	2	o	144.660
21w	p4.4 u1.0	TB	1	i	-7.450
21w	p8.0 u1.0	TB	3	i	-35.707
21w	p8.0 u1.0	TB	1	i	159.167
21w	p8.0 u1.0	TB	2	i	-143.902
21w	p4.2 u1.0	LR	1	i	-94.574
21w	p8.0 u1.0	LR	1	i	-63.642
21w	p8.0 u0.0	training	1	i	-111.209
21w	p4.4 u1.0	TB	1	o	-111.209

21w	p8.0 u1.0	TB	3	o	1.141
21w	p8.0 u1.0	TB	1	o	-91.254
21w	p8.0 u1.0	TB	2	o	-90.200
21w	p4.2 u1.0	LR	1	o	-90.000
21w	p8.0 u1.0	LR	1	o	-91.597
21w	p8.0 u0.0	training	1	o	126.422

A)



B)

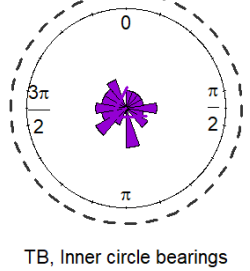
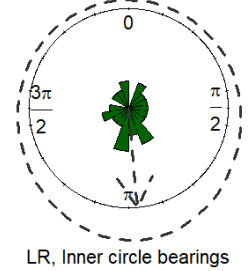


Figure A - Visualization of the raw data of all “LR, inner circle” and “TB, inner circle” trails in a circular plot (A) and representation of the bimodal MLE (B) without the discrimination between strong and weak polarization
A) The feeder is located at the center (cross). Each dot at the edge of the circle represents a measured bearing of the inner circle. The arrow indicates the mean of the direction vector and is proportional to the mean vector length. A shorter arrow refers to a weaker concentration of the data. B) For each condition, a bimodal distribution is suggested. The bars show the density for the estimated distribution. The dashed oval is the confidence interval for the estimate.
The affection of the bumblebee’s ability to use polarization information for orientation within different intensities is not able to see, therefore this representation is not informative.

Eigenständigkeitserklärung

Ich, Marie Jansen, 01/1186437 versichere hiermit, dass ich die anliegende Bachelorarbeit mit dem Thema

“Path integration as a form of home vector navigation with different intensity of visual, artificial polarization pattern cues in walking bumblebees”

selbstständig verfasst und keine anderen Hilfsmittel und Quellen als die angegebenen benutzt habe.

Ich habe textgenerierende KI-Tools als Hilfsmittel verwendet. Mir ist bewusst, dass ich allein für die inhaltliche Richtigkeit der KI, generierten Textpassagen und sonstigen KI-generierten Inhalte sowie für die Kennzeichnung von Formulierungen und Ideen anderer Personen gemäß den Grundsätzen der guten wissenschaftlichen Praxis verantwortlich bin. Ich habe KI-Tools wie unten beschrieben eingesetzt.

- Consensus für die Suche von zugänglichen veröffentlichten Papern
- ChatGPT 4.0 als Hilfe für das Erstellen eines R-Scripts
- ChatGPT 4.0 zum Generieren eines digitalen Bildes des Aufbaus des Experimentes, welches vorher von mir, Marie Jansen, gezeichnet wurde
- DeepL als Hilfestellung zur Synonymfindung und zur Überprüfung der Rechtschreibung und des Satzbaus

Alle Stellen, die anderen – von natürlichen Personen verfassten – Werken (auch aus dem Internet oder anderen elektronischen Text- und Datensammlungen entnommen) dem Wortlaut oder dem Sinn nach entnommen sind, habe ich durch Angabe der Quelle bzw. der Sekundärliteratur entsprechend der Zitierrichtlinien im Fach als Entlehnung kenntlich gemacht.

Ich habe zur Kenntnis genommen, dass die Prüfungsleistung bei Täuschung über die Eigenständigkeit oder die Benutzung nicht zugelassener oder ggf. zugelassener aber nicht ausreichend angegebener Hilfsmittel als „nicht bestanden (5,0)“ bewertet wird und das in besonders schwerwiegenden Täuschungsfällen der zuständige Prüfungsausschuss mich von der Wiederholungsprüfung ausschließen kann mit der Folge des endgültigen Verlustes des Prüfungsanspruchs.

Eine aktuelle Immatrikulationsbescheinigung ist beigelegt.

Konstanz, den 04.08.2025

Datum

Unterschrift Student/in